

LOCAL EXCLUSION OF RESIDUAL TYPE I NAVIER–STOKES SEREGIN LIMITS

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ABSTRACT. We assemble the refined residual analysis for Type I Navier–Stokes Seregin limits. A finite-energy Type I singularity produces, through the Seregin blow-up procedure, a nonzero bounded ancient solution of the centered self-similar Navier–Stokes equation carrying positive local velocity L^3 -mass. After the previously treated small, stationary L^3 , uniformly L^3 -tight, and closed structured/decay alternatives have been removed, the remaining residual class is divided into four analytic classes: axisymmetric bounded-circulation profiles, rotational relative equilibria, degenerate stationary-hull profiles with retained local concentration, and the generic terminal residual.

The first class is excluded by a centered angular-circulation equation and an axis-absorption Liouville theorem. The second class is closed by a local Coriolis-flux identity and a local concentration decomposition: a superthreshold rotational flux gives a nontrivial recentered local profile, while a subthreshold flux remains inside the compact local limiting equation. The third class is reduced, by scale reselection for time-translation hulls, to a stationary profile in a degenerate stationary family which occurs as a Seregin limit and still carries local CKN concentration; it is then excluded by the terminal local concentration theorem. The final class is reduced by covariant tail normalization, exclusion of finite separated concentration profiles, indecomposable-profile sequence- L^3 completeness, mildness inheritance, and the Albritton–Barker endpoint Liouville theorem. The paper distinguishes statements imported from Papers I–III and statements proved here. No global Coriolis–Leray Liouville theorem for bounded profiles or global tail estimate is used in the residual closure.

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1. INTRODUCTION AND MAIN THEOREM

We write the incompressible Navier–Stokes equations on $\mathbb{R}^3 \times (0, T)$ as

$$(1.1) \quad \partial_t u - \Delta u + (u \cdot \nabla)u + \nabla p = 0, \quad \nabla \cdot u = 0.$$

The residual analysis in this paper begins after the standard Type I blow-up construction has been performed. Thus the basic solution is a smooth bounded ancient solution of the centered equation

$$(1.2) \quad \partial_\tau V - \Delta V + \frac{1}{2}y \cdot \nabla V + \frac{1}{2}V + (V \cdot \nabla)V + \nabla P = 0, \quad \nabla \cdot V = 0,$$

on $\mathbb{R}_y^3 \times \mathbb{R}_\tau$. The equation is obtained from the physical ancient representative (U, Q) by

$$(1.3) \quad y = \frac{x}{\sqrt{-t}}, \quad \tau = -\log(-t), \quad V(y, \tau) = \sqrt{-t}U(x, t), \quad P(y, \tau) = (-t)Q(x, t),$$

where $t < 0$ and pressures are always understood modulo addition of a function of time. Conversely,

$$(1.4) \quad U(x, t) = (-t)^{-1/2}V\left(\frac{x}{\sqrt{-t}}, -\log(-t)\right), \quad t < 0.$$

The purpose of the paper is to present the refined residual decomposition as a single standard PDE argument, using only the reductions already established in the preceding papers of the series. To keep the logical status clear, we first state the imported Paper 0 and Paper I–III dependencies. Apart from those four companion papers, the only external inputs used below are the cited peer-reviewed Navier–Stokes regularity and Liouville theorems. All rotational-flux estimates, terminal local concentration decompositions, exclusions of local compactness failure and diffuse exterior concentration, and finite separated-profile exclusions used below are proved in the present paper from local estimates rather than imported as global estimates.

Theorem 1.1 (Paper I Type I Seregin extraction theorem). *A finite-energy Type I singular point produces a normalized Seregin ancient limit (V, P) satisfying (1.2), with*

$$\|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} \leq M < \infty,$$

local suitability and local pressure-gauge compactness, and a compact parabolic cylinder $K \Subset \mathbb{R}^3 \times \mathbb{R}$ for which

$$(1.5) \quad \iint_K |V|^3 dy d\tau \geq \eta_0 > 0.$$

The collection of all normalized Seregin limits generated by the original solution is denoted by $\mathcal{S}$.

Proof. This is Paper I’s extraction theorem [2], including velocity nonvanishing, local suitability, and local pressure compactness. The underlying positive local concentration input comes from Paper 0 [1]. The present paper uses this as an imported reduction and does not reprove the Type I blow-up compactness theorem. $\square$

Definition 1.2 (Imported earlier alternatives). Within a fixed normalized Seregin collection $\mathcal{S}$, the small-amplitude, stationary L^3 , uniformly L^3 -tight, and closed structured/decay classes are the non-residual alternatives defined in Paper I. The small-amplitude and stationary L^3 alternatives are closed in Paper I; the uniformly tight class is excluded in Paper II [3]; and the closed structured/decay class is precisely the part of the structured alternatives proved empty in Paper III [4]. Any structured profile not belonging to that closed Paper I class is residual by definition and is handled by the present paper. The present paper does not reprove those earlier alternatives and does not add any hypothesis to their statements.

Definition 1.3 (Coarse residual class). Let I, J be the index sets used in the base Type I residual criterion. The coarse residual class $\mathcal{R}(\mathcal{S}; I, J)$ is the complement in $\mathcal{S}$ of the already removed alternatives in Theorem 1.2.

Definition 1.4 (Refined residual decomposition). The refined residual decomposition is the ordered decomposition of the remaining coarse residual class into the following alternatives, each understood after all earlier alternatives and all previously closed non-residual classes have been removed:

$$\mathcal{C}_{\text{ax}}, \quad \mathcal{C}_{\text{rot}}, \quad \mathcal{C}_{\text{stat}}^{\text{ret}}, \quad \mathcal{C}_{\text{gen}}.$$

Here $\mathcal{C}_{\text{ax}}$ is the axisymmetric bounded-circulation class, $\mathcal{C}_{\text{rot}}$ is the rotational relative-equilibrium class, $\mathcal{C}_{\text{stat}}^{\text{ret}}$ is the degenerate stationary-hull class with retained local concentration, and $\mathcal{C}_{\text{gen}}$ is the generic terminal residual class.

Theorem 1.5 (Refined residual closure by local concentration analysis). *Relative to the imported Paper I–III alternatives in Theorems 1.1 and 1.2, one has*

$$\mathcal{C}_{\text{ax}} = \emptyset, \quad \mathcal{C}_{\text{rot}} = \emptyset, \quad \mathcal{C}_{\text{stat}}^{\text{ret}} = \emptyset, \quad \mathcal{C}_{\text{gen}} = \emptyset.$$

Consequently the coarse residual class $\mathcal{R}(\mathcal{S}; I, J)$ is empty.

Proof. The axisymmetric bounded-circulation class is excluded by Theorem 3.3. The rotational relative-equilibrium class is excluded by Theorem 4.8, which uses the localized Coriolis-flux identity and a terminal local concentration decomposition rather than a whole-space Coriolis–Leray Liouville theorem. The degenerate stationary-hull class with retained local concentration $\mathcal{C}_{\text{stat}}^{\text{ret}}$ is excluded by Theorem 6.6, using the terminal local concentration theorem proved in Sections 8 and 9. Finally, the generic terminal residual class $\mathcal{C}_{\text{gen}}$ is excluded by Theorem 10.2. These four ordered exclusions exhaust the refined residual decomposition. $\square$

Corollary 1.6 (Paper I residual input). *For every normalized Seregin collection $\mathcal{S}$ generated by a Paper 0-admissible finite-energy Type I singularity, the Paper I remaining class is empty.*

Proof. Paper I defines its remaining class as the complement, inside $\mathcal{S}$, of the small-amplitude, stationary L^3 , uniformly L^3 -tight, and closed structured/decay classes. The notation $\mathcal{R}(\mathcal{S}; I, J)$ used here records the same complement together with the ordered removals used in the residual decomposition; the indices I, J add no analytic assumption. By Theorem 1.5, this class is empty. $\square$

Theorem 1.7 (Type I exclusion after local residual closure). *No Paper 0-admissible finite-energy Type I singularity exists whose normalized Seregin collection is $\mathcal{S}$.*

Proof. Suppose that a Type I singular point exists. By Theorem 1.1, the Seregin blow-up procedure produces a normalized ancient limit (V, P) satisfying (1.2) and the local velocity nonvanishing condition (1.5). By the Paper I class exhaustion, V is either in one of the non-residual classes recorded in Theorem 1.2 or in $\mathcal{R}(\mathcal{S}; I, J)$. The first alternative is impossible by the corresponding closed inputs: the small and stationary L^3 classes are excluded in Paper I,

the uniformly L^3 -tight class is excluded in Paper II, and the closed structured/decay class is excluded in Paper III. The second alternative is impossible by [Theorem 1.5](#). This contradiction proves the theorem. $\square$

2. SHARED LOCAL NOTATION

For $z_0 = (y_0, \tau_0)$ and $r > 0$, write

$$Q_r(z_0) := B_r(y_0) \times (\tau_0 - r^2, \tau_0).$$

For a centered profile (V, P) , the pressure-normalized local critical quantity is

$$(2.1) \quad \mathcal{E}_r(V, P; z_0) := \iint_{Q_r(z_0)} \left(|V|^3 + |P - c_{r,z_0}(\tau)|^{3/2} \right) dy d\tau,$$

where $c_{r,z_0}(\tau)$ is an allowed time-dependent pressure gauge, usually chosen as a spatial average over $B_r(y_0)$. All local pressure convergences below are understood after such gauges.

Definition 2.1 (Retained local concentration profile). Fix $\varepsilon_* > 0$. A terminal profile (U, Π) has retained local concentration if, after the terminal normalization under consideration,

$$\mathcal{E}_1(U, \Pi; 0) \geq \varepsilon_*.$$

Definition 2.2 (Refined residual candidate). A refined residual candidate is a bounded normalized Seregin ancient limit obtained from the Paper I extraction theorem which carries the positive compact local velocity mass in (1.5) and which is not in any alternative already closed before the refined residual decomposition is entered. The definition is local: the only persistent lower bound attached to the candidate by Paper I is the fixed compact velocity L^3 lower bound.

Remark 2.3 (No hidden global L^3 assertion). The terminal arguments below do not claim that a generic residual profile is globally L^3 -tight or globally bounded in $L^\infty_\tau L^3_y$. The only nonlocal critical-norm conclusion used at the endpoint is the weaker sequence- L^3 completeness of retained indecomposable concentration profiles, and that conclusion is derived from local terminal alternatives rather than from an a priori global estimate on the original residual class.

3. EXCLUSION OF THE AXISYMMETRIC BOUNDED-CIRCULATION RESIDUAL CLASS

In this section we prove the rigidity statement needed for the first refined residual class. The argument uses the scale-invariant angular circulation in centered variables. The essential point is that the centered Type I equation contains the Ornstein–Uhlenbeck drift. Thus the angular-circulation equation has an absorbing boundary at the symmetry axis together with a confining drift at spatial infinity. This gives a scalar Liouville theorem for bounded ancient solutions of the circulation equation.

Throughout the section we write cylindrical coordinates as (ρ, θ, Z) , and an axisymmetric vector field as

$$V = V_\rho e_\rho + V_\theta e_\theta + V_Z e_Z.$$

The centered Navier–Stokes equation is

$$(3.1) \quad \partial_\tau V - \Delta V + \frac{1}{2}y \cdot \nabla V + \frac{1}{2}V + (V \cdot \nabla)V + \nabla P = 0, \quad \nabla \cdot V = 0.$$

For an axisymmetric solution define the centered angular circulation

$$(3.2) \quad \Gamma(\rho, Z, \tau) := \rho V_\theta(\rho, Z, \tau).$$

This is the same scale-invariant quantity as the physical circulation rU_θ . If the physical representative is

$$U(x, t) = (-t)^{-1/2} V(x/\sqrt{-t}, -\log(-t)),$$

then

$$rU_\theta(r, z, t) = \rho V_\theta(\rho, Z, \tau), \quad r = \sqrt{-t} \rho, \quad z = \sqrt{-t} Z.$$

Lemma 3.1 (Centered circulation equation). *Let (V, P) be a smooth bounded axisymmetric solution of (3.1). Then $\Gamma = \rho V_\theta$ satisfies*

$$(3.3) \quad \partial_\tau \Gamma = \partial_\rho^2 \Gamma - \frac{1}{\rho} \partial_\rho \Gamma + \partial_Z^2 \Gamma - \left(V_\rho + \frac{\rho}{2}\right) \partial_\rho \Gamma - \left(V_Z + \frac{Z}{2}\right) \partial_Z \Gamma$$

on $(0, \infty) \times \mathbb{R} \times \mathbb{R}$. Moreover

$$(3.4) \quad \Gamma(0, Z, \tau) = 0 \quad \text{for all } (Z, \tau) \in \mathbb{R} \times \mathbb{R}.$$

Proof. The angular component of (3.1) is

$$\partial_\tau V_\theta - \left(\Delta - \frac{1}{\rho^2}\right) V_\theta + \frac{1}{2}(\rho \partial_\rho + Z \partial_Z) V_\theta + \frac{1}{2} V_\theta + V_\rho \partial_\rho V_\theta + V_Z \partial_Z V_\theta + \frac{V_\rho V_\theta}{\rho} = 0.$$

Multiplying by ρ , using

$$\begin{aligned} \rho \left(\Delta - \frac{1}{\rho^2}\right) V_\theta &= \partial_\rho^2 \Gamma - \frac{1}{\rho} \partial_\rho \Gamma + \partial_Z^2 \Gamma, \\ \rho \left[\frac{1}{2}(\rho \partial_\rho + Z \partial_Z) V_\theta + \frac{1}{2} V_\theta\right] &= \frac{1}{2}(\rho \partial_\rho + Z \partial_Z) \Gamma, \end{aligned}$$

and

$$\rho \left(V_\rho \partial_\rho V_\theta + V_Z \partial_Z V_\theta + \frac{V_\rho V_\theta}{\rho}\right) = V_\rho \partial_\rho \Gamma + V_Z \partial_Z \Gamma,$$

gives (3.3). Finally, smoothness of a bounded axisymmetric vector field at the axis implies $V_\theta(\rho, Z, \tau) = O(\rho)$ as $\rho \downarrow 0$, locally uniformly in (Z, τ) . Hence $\Gamma = \rho V_\theta = O(\rho^2)$, which gives (3.4). $\square$

We next record the scalar Liouville theorem used below. It is stated in a form that isolates precisely the PDE mechanism: bounded drift perturbations of the centered circulation operator cannot sustain a bounded ancient solution which vanishes on the symmetry axis.

Lemma 3.2 (Axis-absorption Liouville theorem). *Let $B = (B_\rho, B_Z)$ be a smooth vector field on $(0, \infty) \times \mathbb{R} \times \mathbb{R}$ satisfying*

$$\|B\|_{L^\infty} < \infty.$$

Let $G \in C^{2,1}((0, \infty) \times \mathbb{R} \times \mathbb{R}) \cap L^\infty$ solve

$$(3.5) \quad \partial_\tau G = \partial_\rho^2 G - \frac{1}{\rho} \partial_\rho G + \partial_Z^2 G - \left(B_\rho + \frac{\rho}{2}\right) \partial_\rho G - \left(B_Z + \frac{Z}{2}\right) \partial_Z G$$

for all $\tau \in \mathbb{R}$, and assume

$$(3.6) \quad \lim_{\rho \downarrow 0} G(\rho, Z, \tau) = 0 \quad \text{locally uniformly in } (Z, \tau).$$

Then $G \equiv 0$.

Proof. Set $M_B = \|B\|_{L^\infty}$. For $R > 1$ define

$$(3.7) \quad h_R(\rho) := \frac{\int_0^{\min\{\rho, R\}} s \exp\left(\frac{s^2}{4} - M_B s\right) ds}{\int_0^R s \exp\left(\frac{s^2}{4} - M_B s\right) ds}.$$

Then $0 \leq h_R \leq 1$, $h_R(0) = 0$, $h_R(R) = 1$, and, on $0 < \rho < R$,

$$(3.8) \quad h_R'' + \left(M_B - \frac{\rho}{2} - \frac{1}{\rho} \right) h_R' = 0, \quad h_R' > 0.$$

Let

$$\mathcal{L}_\tau f := \partial_\rho^2 f - \frac{1}{\rho} \partial_\rho f + \partial_Z^2 f - \left(B_\rho + \frac{\rho}{2} \right) \partial_\rho f - \left(B_Z + \frac{Z}{2} \right) \partial_Z f.$$

Since $B_\rho \geq -M_B$, (3.8) gives

$$(3.9) \quad \mathcal{L}_\tau h_R = h_R'' - \left(\frac{1}{\rho} + \frac{\rho}{2} + B_\rho \right) h_R' \leq h_R'' + \left(M_B - \frac{\rho}{2} - \frac{1}{\rho} \right) h_R' = 0.$$

Thus $(\partial_\tau - \mathcal{L}_\tau)h_R \geq 0$.

We now compare G to this barrier on the strip $\Omega_R = (0, R) \times \mathbb{R}$. Let $M_G = \|G\|_{L^\infty}$. We first record the killed-strip decay estimate used by the comparison argument. If $q_{s,R}$ solves

$$\partial_\tau q_{s,R} = \mathcal{L}_\tau q_{s,R}, \quad q_{s,R} = 0 \text{ on } \rho = 0, R, \quad q_{s,R}(\rho, Z, s) = 1,$$

then, for every compact $K \Subset \Omega_R$ and every fixed t ,

$$(3.10) \quad \lim_{s \rightarrow -\infty} \sup_{(\rho, Z) \in K} q_{s,R}(\rho, Z, t) = 0.$$

To prove (3.10), fix $K \Subset \Omega_R$. Choose $\delta > 0$, $Z_K > 0$, and $Z_1 > Z_K + 1$ so that

$$K \subset [\delta, R - \delta] \times [-Z_K, Z_K] \subset (0, R) \times (-Z_1, Z_1).$$

On the finite cylinder

$$\Omega_{R,Z_1} := (0, R) \times (-Z_1, Z_1)$$

with Dirichlet data on the full parabolic boundary, the operator $\mathcal{L}_\tau$ is uniformly parabolic on $[\delta/2, R - \delta/2] \times [-Z_1, Z_1]$, has bounded first-order coefficients there, and has absorbing boundary at $\rho = 0, R$. The boundary parabolic Harnack inequality and the strong maximum principle give a time $T_R > 0$ and a number $\vartheta_R \in (0, 1)$, depending only on R, M_B, δ, Z_K and not on Z_1 , such that every nonnegative solution with initial data bounded by 1 and zero lateral data satisfies

$$0 \leq q(\cdot, s + T_R) \leq 1 - \vartheta_R \quad \text{on } [\delta, R - \delta] \times [-Z_K, Z_K].$$

Applying the same estimate after each time translation and using the maximum principle between the compact set and the boundary gives

$$\sup_K q_{s,R}(\cdot, s + mT_R) \leq (1 - \vartheta_R)^m, \quad m = 1, 2, \dots$$

For a fixed $t > s$, choose m with $s + mT_R \leq t < s + (m+1)T_R$. The maximum principle and interior Harnack estimate on the final interval $[s + mT_R, t]$ preserve the preceding bound up to a constant depending only on R, M_B, δ, Z_K, T_R . Letting $s \rightarrow -\infty$ forces $m \rightarrow \infty$, proving decay on K for the truncated problem. The solutions on Ω_{R,Z_1} increase monotonically to the killed-strip solution as $Z_1 \rightarrow \infty$; passing to this limit gives (3.10).

Fix $R > 1$, $s < t$, and set

$$W_R^+ := \frac{G}{M_G} - h_R, \quad W_R^- := -\frac{G}{M_G} - h_R,$$

with the convention that the conclusion is immediate if $M_G = 0$. By (3.5) and (3.9),

$$(\partial_\tau - \mathcal{L}_\tau)W_R^\pm \leq 0 \quad \text{in } \Omega_R \times (s, t).$$

On $\rho = 0$, (3.6) and $h_R(0) = 0$ give $W_R^\pm \leq 0$. On $\rho = R$, $h_R(R) = 1$ and $|G| \leq M_G$ give again $W_R^\pm \leq 0$. The parabolic comparison principle with the killed-strip kernel therefore yields, for $\tau = t$,

$$(W_R^\pm)_+(\rho, Z, t) \leq q_{s,R}(\rho, Z, t) \quad ((\rho, Z) \in \Omega_R).$$

Letting $s \rightarrow -\infty$ and using (3.10) gives $W_R^\pm \leq 0$ on compact subsets of Ω_R . Since the compact subset is arbitrary,

$$(3.11) \quad |G(\rho, Z, t)| \leq M_G h_R(\rho) \quad (0 < \rho < R, Z \in \mathbb{R}, t \in \mathbb{R}).$$

Finally, for each fixed $\rho < \infty$, the denominator in (3.7) tends to infinity super-exponentially as $R \rightarrow \infty$, while the numerator remains fixed. Hence $h_R(\rho) \rightarrow 0$. Letting $R \rightarrow \infty$ in (3.11) gives $G(\rho, Z, t) = 0$ for every $\rho > 0$, $Z \in \mathbb{R}$, and $t \in \mathbb{R}$. The boundary value at $\rho = 0$ is already zero by assumption. Therefore $G \equiv 0$. $\square$

Theorem 3.3 (Exclusion of the axisymmetric bounded-circulation residual class). *Let $\mathcal{C}_{\text{ax}}(\mathcal{S}; I, J)$ be the axisymmetric bounded-circulation residual class in the refined residual decomposition. Then*

$$(3.12) \quad \mathcal{C}_{\text{ax}}(\mathcal{S}; I, J) = \emptyset.$$

Proof. Assume, for contradiction, that $V \in \mathcal{C}_{\text{ax}}(\mathcal{S}; I, J)$. By definition of $\mathcal{C}_{\text{ax}}$, the associated physical ancient representative is axisymmetric and has bounded angular circulation. Equivalently, in centered variables,

$$\Gamma(\rho, Z, \tau) = \rho V_\theta(\rho, Z, \tau)$$

is bounded on $(0, \infty) \times \mathbb{R} \times \mathbb{R}$. Since V is a Seregin ancient limit, it is smooth and bounded in centered variables. Hence $B = (V_\rho, V_Z)$ is bounded.

By Theorem 3.1, Γ satisfies (3.3) and vanishes at the axis. Applying Theorem 3.2 with $G = \Gamma$ and $B = (V_\rho, V_Z)$ gives

$$\Gamma \equiv 0.$$

Therefore $V_\theta \equiv 0$ for $\rho > 0$, and by smoothness also on the axis. Thus the associated ancient solution is axisymmetric without swirl. But the axisymmetric no-swirl class is one of the structured alternatives removed before the residual class is defined. This contradicts $V \in \mathcal{R}(\mathcal{S}; I, J)$, and therefore no such V exists. $\square$

Remark 3.4 (Why this does not assert the full bounded-swirl Liouville theorem). The proof uses the centered Type I equation, specifically the Ornstein–Uhlenbeck drift terms $-\rho \partial_\rho/2$ and $-Z \partial_Z/2$ in the scalar equation for Γ . It is therefore a Type I Seregin limit argument. It does not prove that every bounded ancient axisymmetric solution of the unrenormalized Navier–Stokes equations with bounded physical circulation is swirl-free. The latter lacks the confining centered drift and remains a different shell-transport problem.

4. THE ROTATIONAL RELATIVE-EQUILIBRIUM RESIDUAL CLASS

We next isolate the second refined residual class. This class consists of centered Type I ancient limits whose time dependence is a rigid rotation of a fixed spatial profile. Passing to the co-rotating frame converts the centered Navier–Stokes equation into a stationary Leray profile equation with a Coriolis transport term. A whole-space Coriolis–Leray Liouville theorem for all bounded profiles is not used here. Instead, the Coriolis term is localized as an annular flux and is then treated by the local concentration analysis proved in Sections 8 and 9.

Throughout this section the centered equation is

$$(4.1) \quad \partial_\tau V - \Delta V + \frac{1}{2} y \cdot \nabla V + \frac{1}{2} V + (V \cdot \nabla) V + \nabla P = 0, \quad \nabla \cdot V = 0.$$

For $\Omega \in \mathfrak{so}(3)$, set

$$(4.2) \quad Q(\tau) := \exp(\tau\Omega) \in SO(3).$$

Definition 4.1 (Rotational relative-equilibrium profile). A bounded smooth ancient solution (V, P) of (4.1) is a rotational relative equilibrium if there exist $\Omega \in \mathfrak{so}(3)$, a bounded smooth divergence-free vector field W , and a pressure profile Π , such that

$$(4.3) \quad V(y, \tau) = Q(\tau)W(Q(\tau)^T y), \quad P(y, \tau) = \Pi(Q(\tau)^T y)$$

up to the usual time-dependent pressure gauge. The class $\mathcal{C}_{\text{rot}}$ consists of such residual Seregin profiles, after the axisymmetric bounded-circulation class and the previously closed Paper I–III alternatives have been removed.

Lemma 4.2 (Co-rotating stationary equation). *Let (V, P) be a rotational relative equilibrium. Then the co-rotating profile (W, Π) satisfies*

$$(4.4) \quad -\Delta W + \frac{1}{2}y \cdot \nabla W + \frac{1}{2}W + (W \cdot \nabla)W + \Omega W - (\Omega y) \cdot \nabla W + \nabla \Pi = 0, \quad \nabla \cdot W = 0.$$

Equivalently,

$$(4.5) \quad \mathbb{P} \left[-\Delta W + \frac{1}{2}y \cdot \nabla W + \frac{1}{2}W + (W \cdot \nabla)W + \Omega W - (\Omega y) \cdot \nabla W \right] = 0,$$

where $\mathbb{P}$ is the whole-space Leray projector.

Proof. Let $z = Q(\tau)^T y$. Since $Q'(\tau) = \Omega Q(\tau)$ and $Q(\tau)^T \Omega Q(\tau) = \Omega$, one has

$$\partial_\tau z = -\Omega z.$$

Therefore

$$\partial_\tau V(y, \tau) = Q(\tau) \left(\Omega W(z) - (\Omega z) \cdot \nabla W(z) \right).$$

The Laplacian, divergence, convection term, and pressure gradient are invariant under the orthogonal change of variables. Substitution into (4.1), followed by multiplication by $Q(\tau)^T$, gives (4.4). Applying the Leray projector gives (4.5). $\square$

4.1. The Bernoulli identity and the local Coriolis flux. Write

$$(4.6) \quad F(y) := \frac{1}{2}y - \Omega y, \quad A(y) := W(y) + F(y), \quad \omega := \nabla \times W.$$

Since Ω is skew-symmetric, there is a unique vector $\varpi \in \mathbb{R}^3$ such that

$$(4.7) \quad \Omega y = \varpi \times y.$$

Then

$$(4.8) \quad \nabla \times F = -2\varpi.$$

Lemma 4.3 (Coriolis Bernoulli identity). *Let (W, Π) be a smooth solution of (4.4). Define*

$$(4.9) \quad B := \frac{1}{2}|W|^2 + \Pi + F \cdot W.$$

Then

$$(4.10) \quad \nabla B + \nabla \times \omega - A \times \omega = 0,$$

and consequently

$$(4.11) \quad -\Delta B + A \cdot \nabla B = -|\omega|^2 - \omega \cdot (\nabla \times F) = -|\omega|^2 + 2\varpi \cdot \omega.$$

Proof. The vector identity

$$(W \cdot \nabla)W = \nabla \left(\frac{1}{2}|W|^2 \right) - W \times \omega$$

and the identity

$$(F \cdot \nabla)W + (\nabla F)^T W = \nabla(F \cdot W) - F \times \omega$$

apply because $(\nabla F)^T W = \frac{1}{2}W + \Omega W$. Thus (4.4) can be rewritten as

$$-\Delta W + \nabla \left(\frac{1}{2}|W|^2 + \Pi + F \cdot W \right) - (W + F) \times \omega = 0.$$

Since $\nabla \cdot W = 0$, one has $-\Delta W = \nabla \times \omega$, which gives (4.10).

Taking divergence in (4.10) gives

$$\Delta B = \nabla \cdot (A \times \omega),$$

because the divergence of a curl vanishes. Using

$$\nabla \cdot (A \times \omega) = \omega \cdot (\nabla \times A) - A \cdot (\nabla \times \omega), \quad \nabla \times A = \omega + \nabla \times F,$$

we obtain

$$\Delta B = |\omega|^2 + \omega \cdot (\nabla \times F) - A \cdot (\nabla \times \omega).$$

On the other hand, by (4.10),

$$A \cdot \nabla B = A \cdot (A \times \omega) - A \cdot (\nabla \times \omega) = -A \cdot (\nabla \times \omega).$$

Combining the last two identities gives (4.11). Finally, (4.8) follows from (4.7). $\square$

Lemma 4.4 (Localized Coriolis flux identity). *Let (W, Π) be as in Theorem 4.3, and let $\chi \in C_c^\infty(\mathbb{R}^3)$. Then*

$$(4.12) \quad \int_{\mathbb{R}^3} \chi (-\Delta B + A \cdot \nabla B + |\omega|^2) dy = -2 \int_{\mathbb{R}^3} (W \times \varpi) \cdot \nabla \chi dy.$$

In particular the Coriolis contribution is supported only where the cutoff varies.

Proof. By (4.11),

$$-\Delta B + A \cdot \nabla B + |\omega|^2 = 2\varpi \cdot \omega.$$

Because ϖ is constant and $\omega = \nabla \times W$,

$$\varpi \cdot \omega = \nabla \cdot (W \times \varpi).$$

Multiplying by χ , integrating over $\mathbb{R}^3$, and integrating by parts gives

$$2 \int \chi \varpi \cdot \omega = 2 \int \chi \nabla \cdot (W \times \varpi) = -2 \int (W \times \varpi) \cdot \nabla \chi,$$

which proves (4.12). $\square$

Lemma 4.5 (Non-small Coriolis flux creates a retained local concentration). *Let $\chi \in C_c^\infty(\mathbb{R}^3)$, let $A_\chi := \text{supp } \nabla \chi$, and put*

$$C_\chi := \int_{A_\chi} |\nabla \chi|^{3/2} dy.$$

Fix $T \in (0, 1]$, and let

$$\mathcal{T}_{\chi, T} := \{(y, \tau) : -T < \tau < 0, Q(\tau)^T y \in A_\chi\}$$

be the swept annular tube in the original centered variables. Cover $\mathcal{T}_{\chi,T}$ by finitely many unit terminal cylinders

$$\mathcal{T}_{\chi,T} \subset \bigcup_{m=1}^{N_{\chi,T}} Q_1(z_m).$$

Let $\varepsilon_{\text{sd}} > 0$ be the local concentration threshold used in [Theorem 8.2](#). If $\varpi \neq 0$ and

$$(4.13) \quad \left| 2 \int_{\mathbb{R}^3} (W \times \varpi) \cdot \nabla \chi \, dy \right| \geq 2|\varpi| C_{\chi}^{2/3} \left(\frac{N_{\chi,T} \varepsilon_{\text{sd}}}{T} \right)^{1/3},$$

then at least one cylinder $Q_1(z_m)$ is ε_{sd} -concentrating for the original centered solution V .

Proof. Write

$$\delta_{\chi} := \left| 2 \int_{\mathbb{R}^3} (W \times \varpi) \cdot \nabla \chi \, dy \right|.$$

Holder's inequality gives

$$\begin{aligned} \delta_{\chi} &\leq 2|\varpi| \int_{A_{\chi}} |W| |\nabla \chi| \, dy \\ &\leq 2|\varpi| \left(\int_{A_{\chi}} |W|^3 \, dy \right)^{1/3} C_{\chi}^{2/3}. \end{aligned}$$

Thus (4.13) implies

$$\int_{A_{\chi}} |W|^3 \, dy \geq \frac{N_{\chi,T} \varepsilon_{\text{sd}}}{T}.$$

Since $V(y, \tau) = Q(\tau)W(Q(\tau)^T y)$ and $Q(\tau)$ is an isometry,

$$\begin{aligned} \iint_{\mathcal{T}_{\chi,T}} |V(y, \tau)|^3 \, dy \, d\tau &= \int_{-T}^0 \int_{Q(\tau)A_{\chi}} |W(Q(\tau)^T y)|^3 \, dy \, d\tau \\ &= T \int_{A_{\chi}} |W(z)|^3 \, dz \geq N_{\chi,T} \varepsilon_{\text{sd}}. \end{aligned}$$

Because the tube is covered by $N_{\chi,T}$ unit terminal cylinders, one of them satisfies

$$\iint_{Q_1(z_m)} |V|^3 \, dy \, d\tau \geq \varepsilon_{\text{sd}}.$$

The pressure contribution to the local CKN measure is nonnegative after the allowed pressure gauge. Hence

$$\mu(Q_1(z_m)) \geq \varepsilon_{\text{sd}},$$

so the corresponding recentering sequence is ε_{sd} -concentrating in the sense of [Theorem 8.2](#). $\square$

Lemma 4.6 (Local Coriolis-flux dichotomy). *Fix $\chi \in C_c^{\infty}(\mathbb{R}^3)$ and $T \in (0, 1]$. Along every terminal extraction of a rotational relative equilibrium, exactly one of the following local alternatives occurs:*

- (i) $\varpi \neq 0$, the flux across $A_{\chi} = \text{supp } \nabla \chi$ satisfies the threshold (4.13), and the swept annular tube contains a unit-scale concentration;
- (ii) either $\varpi = 0$, or the threshold (4.13) fails; then the flux is a bounded compactly supported functional of W on A_{χ} , passes to terminal limits by local L^3 convergence, and creates no terminal alternative other than the single-profile, finite separated-profile, exterior-escape, diffuse-exterior-concentration, local-vanishing, or local-compactness-failure alternatives of [Theorem 8.6](#).

Proof. Alternative (i) is precisely [Theorem 4.5](#). If $\varpi = 0$, then the flux vanishes identically. If $\varpi \neq 0$ and (4.13) fails, the flux is bounded by the right hand side of that threshold. In both cases the functional

$$W \mapsto -2 \int_{\mathbb{R}^3} (W \times \varpi) \cdot \nabla \chi \, dy$$

depends only on W on the compact annular set A_χ . Under the local Seregin compactness supplied by Paper I, velocities converge strongly in L^3_{loc} ; since $\nabla \chi \in L^{3/2}$ and ϖ is fixed, the functional is continuous along every terminal subsequence:

$$\int (W_n \times \varpi) \cdot \nabla \chi \longrightarrow \int (W_\infty \times \varpi) \cdot \nabla \chi.$$

Thus a subthreshold flux is retained only as a local term in the limiting localized Bernoulli identity (4.12). It is not a new location of local velocity mass. Every actual location of compact velocity L^3 -mass is still one of the terminal local concentration alternatives listed in [Theorem 8.6](#). $\square$

Remark 4.7 (Why this is not a global Liouville argument). For $\Omega = 0$, the Coriolis flux in [Theorem 4.4](#) vanishes. For $\Omega \neq 0$, the same lemma does not give a sign-definite maximum principle for B . It gives only the local dichotomy in [Theorem 4.6](#): either a chosen annulus already contains a recentered local concentration, or the annular flux is a subthreshold compact functional which passes to the local terminal limit. No tail decay, whole-space L^p bound, Gaussian energy estimate, or global pressure normalization is used.

4.2. Local closure of the rotational relative-equilibrium class.

Theorem 4.8 (Stratified exclusion of rotational relative equilibria). *The ordered rotational relative-equilibrium residual class is empty:*

$$(4.14) \quad \mathcal{C}_{\text{rot}}(\mathcal{S}; I, J) = \emptyset.$$

Proof. Suppose, for contradiction, that $V \in \mathcal{C}_{\text{rot}}(\mathcal{S}; I, J)$. By [Theorem 4.1](#) and [Theorem 4.2](#), there are $\Omega \in \mathfrak{so}(3)$ and a bounded smooth divergence-free profile (W, Π) satisfying the Coriolis–Leray equation (4.4). The centered representative V is a normalized Seregin ancient limit, so by [Theorem 1.1](#) it carries a fixed positive compact velocity L^3 -mass.

Choose a countable family of smooth cutoffs which is cofinal among compact annuli in the terminal variables. For each cutoff and each rational $T \in (0, 1]$, apply the local dichotomy [Theorem 4.6](#). A superthreshold Coriolis flux produces a unit-scale concentration and is therefore already part of the local concentration decomposition. A subthreshold Coriolis flux is a compactly supported functional of W , continuous under the local convergence supplied by Paper I, and hence remains inside the localized limiting equation; it carries no independent CKN mass and creates no additional terminal alternative.

We now apply the terminal local concentration decomposition to the inherited compact density. Loss of local compactness, local vanishing, exterior escape, and diffuse exterior concentration cannot be the sole locations of this density by [Theorem 8.9](#). By the preceding paragraph, the Coriolis flux has already led either to a local concentration or to a compact local term in the selected terminal equation. Thus, after local concentration extraction, the retained density is located either in a finite separated family of retained concentration profiles or in a single retained concentration terminal profile.

The finite separated-family alternative is impossible by [Theorem 9.13](#). In the single-profile case, [Theorem 9.3](#) makes the terminal profile terminally indecomposable, and [Theorem 9.7](#) excludes retained indecomposable concentration profiles. Both possible locations of the inherited compact density are impossible. This contradicts the normalization of V , and therefore $\mathcal{C}_{\text{rot}}(\mathcal{S}; I, J) = \emptyset$. $\square$

Proposition 4.9 (Already-closed integrable subcase). *Let V be a rotational relative equilibrium with co-rotating profile W . If*

$$(4.15) \quad W \in L^3(\mathbb{R}^3),$$

then $V \equiv 0$. More generally, the same conclusion holds under any condition which places the physical representative in $L_t^\infty L_x^3$ on its ancient time interval.

Proof. Rotations and the Navier–Stokes critical scaling preserve the L^3 norm. Thus $W \in L^3$ implies

$$\sup_{\tau \in \mathbb{R}} \|V(\cdot, \tau)\|_{L^3(\mathbb{R}^3)} < \infty.$$

Equivalently, the physical representative lies in the critical class $L_t^\infty L_x^3$. The endpoint L^3 class is one of the already closed alternatives in Paper I, by the Escauriaza–Seregin–Šverák endpoint regularity theorem [7] and the stationary Nečas–Růžička–Šverák Liouville theorem [11]. Therefore $V \equiv 0$. A zero profile has zero local velocity mass, so no such profile can belong to the residual nonvanishing class. $\square$

5. THE DEGENERATE STATIONARY-HULL RESIDUAL CLASS

We now isolate the third refined residual class. This class is designed to capture Type I ancient limits whose time-translation hull contains a stationary profile with retained local concentration lying on a non-isolated family of stationary centered profiles. The role of the stationary-family condition is to separate genuinely degenerate stationary profile families from isolated stationary profiles and from the symmetry-generated classes already removed in $\mathcal{C}_{\text{ax}}$ and $\mathcal{C}_{\text{rot}}$.

A small point of formulation is important. It is not enough to assume that the hull of a residual ancient solution contains some stationary-family element. A stationary element may occur as a limit without retained local concentration and therefore need not retain the local CKN concentration inherited from the original singular point. Such a hull element should not be placed in a class that is to be closed by a stationary-profile exclusion. Therefore the class below is defined using *retained-concentration occurrence*: the stationary-family profile must itself occur as a Seregin limit carrying a positive local CKN concentration.

Throughout this section the centered equation is

$$(5.1) \quad \partial_\tau V - \Delta V + \frac{1}{2}y \cdot \nabla V + \frac{1}{2}V + (V \cdot \nabla)V + \nabla P = 0, \quad \nabla \cdot V = 0.$$

Its projected form is

$$(5.2) \quad \partial_\tau V = \mathcal{F}(V), \quad \mathcal{F}(W) := \mathbb{P} \left[\Delta W - \frac{1}{2}(W + y \cdot \nabla W) - (W \cdot \nabla)W \right],$$

where $\mathbb{P}$ is the whole-space Leray projector.

5.1. Stationary families and retained-concentration hull limits.

Definition 5.1 (Admissible stationary phase space). An admissible stationary phase space is a Banach space $\mathfrak{B}$ of solenoidal vector fields on $\mathbb{R}^3$ such that:

- (i) $\mathfrak{B} \hookrightarrow C_{\text{loc}}^{2,\alpha}(\mathbb{R}^3) \cap L^\infty(\mathbb{R}^3)$ continuously for some $\alpha \in (0, 1)$;
- (ii) the map $\mathcal{F}$ in (5.2) is well-defined on an open subset $\mathcal{U} \subset \mathfrak{B}$ and maps $\mathcal{U}$ into a Banach space $\mathfrak{V}$;
- (iii) $\mathcal{F} : \mathcal{U} \rightarrow \mathfrak{V}$ is C^1 .

Definition 5.2 (Nontrivial stationary family). Let $W \in \mathcal{U} \subset \mathfrak{B}$ satisfy $\mathcal{F}(W) = 0$. We say that W lies on an admissible nontrivial stationary family if there exist $\varepsilon > 0$ and a C^1 curve

$$a \mapsto W_a \in \mathcal{U}, \quad |a| < \varepsilon,$$

such that

$$(5.3) \quad W_0 = W, \quad \mathcal{F}(W_a) = 0 \quad (|a| < \varepsilon),$$

and such that the family is not locally generated only by the action of the rigid rotation group. Equivalently, after possibly reparametrizing the curve, one may assume that

$$(5.4) \quad \dot{W}_0 \notin \{\Omega W - (\Omega y) \cdot \nabla W : \Omega \in \mathfrak{so}(3)\}$$

whenever the tangent $\dot{W}_0$ exists and is nonzero.

Definition 5.3 (Retained local CKN concentration). Let (W, Π) be a stationary centered profile. We say that (W, Π) is ε_* -concentrating if there exist a compact parabolic cylinder $Q_r(z_0)$ and an admissible pressure gauge c_{r,z_0} such that

$$(5.5) \quad \iint_{Q_r(z_0)} (|W|^3 + |\Pi - c_{r,z_0}|^{3/2}) \, dy \, d\tau \geq \varepsilon_*.$$

For a stationary profile the integrand is independent of τ , but the parabolic-cylinder notation is retained to match the CKN normalization used for Seregin limits.

Definition 5.4 (Degenerate stationary-hull residual class with retained concentration). Fix an admissible stationary phase space $\mathfrak{B}$. The degenerate stationary-hull residual class with retained local concentration

$$\mathcal{C}_{\text{stat}}^{\text{ret}}(\mathcal{S}; I, J; \mathfrak{B})$$

consists of those $V \in \mathcal{R}(\mathcal{S}; I, J)$, after the ordered removal of $\mathcal{C}_{\text{ax}}$ and $\mathcal{C}_{\text{rot}}$, for which there exists a sequence $s_k \in \mathbb{R}$ and a stationary profile (W, Π) such that

$$(5.6) \quad (V(\cdot, \cdot + s_k), P(\cdot, \cdot + s_k) - c_k(\cdot)) \longrightarrow (W, \Pi)$$

locally smoothly for the velocity and locally in $L^{3/2}$ for the pressure modulo admissible gauges, and such that:

- (i) $W \in \mathfrak{B}$ and $\mathcal{F}(W) = 0$;
- (ii) W lies on an admissible nontrivial stationary family in the sense of [Theorem 5.2](#);
- (iii) (W, Π) is ε_* -concentrating in the sense of [Theorem 5.3](#);
- (iv) (W, Π) is not in any previously closed stationary, structured, axisymmetric bounded-circulation, or rotational relative-equilibrium class.

In the ordered residual decomposition one keeps this class of stationary-hull limits with retained concentration as the degenerate stationary-hull alternative. The hull cases without retained local concentration should be assigned to the generic terminal residual class, where they are handled by the terminal local concentration decomposition rather than by a stationary-profile exclusion.

Remark 5.5 (Why retained concentration is necessary). The condition (5.5) prevents the following logical error. A nonstationary ancient solution may have a stationary profile in its hull with zero local CKN mass on the relevant compact cylinders. Excluding that stationary profile would not exclude the original ancient solution. The stationary-profile exclusion used below applies only to stationary profiles that themselves occur as nontrivial Seregin limits with positive local concentration.

5.2. Hull limits are again Seregin limits. The next lemma is the basic compatibility between time-translation hulls in centered variables and rescaling sequences in physical variables. It is the reason that a stationary profile in the hull which retains local concentration may be treated as a blow-up profile of the original solution, not merely as a limit inside the already extracted ancient solution.

Lemma 5.6 (Scale reselection for centered hull limits). *Let (V, P) be a centered Type I Seregin limit obtained from a finite-energy suitable weak solution (u, p) at a singular point $z_* = (x_*, t_*)$. Suppose that, for some sequence $s_k \in \mathbb{R}$,*

$$(V(\cdot, \cdot + s_k), P(\cdot, \cdot + s_k) - c_k(\cdot)) \rightarrow (W, \Pi)$$

locally in the Seregin topology. Then, after passing to a diagonal subsequence and reselecting the physical blow-up scales, (W, Π) is itself a centered Type I Seregin limit of (u, p) at z_ . If the convergence to (W, Π) retains a positive local CKN lower bound, then (W, Π) is a nontrivial Seregin limit with the same type of inherited nonvanishing condition.*

Proof. Let $\lambda_n \downarrow 0$ be a sequence of scales producing (V, P) . In centered variables, a time translation by s is exactly a change of the physical blow-up scale by a factor depending only on s . With the convention used in (5.1), write this adjusted scale as

$$(5.7) \quad \lambda_{n,s} := e^{-s/2} \lambda_n.$$

The rescaled fields built from (u, p) at scale $\lambda_{n,s}$ are the same as the fields at scale λ_n , translated by s in the centered time variable, up to the corresponding pressure gauge.

Fix an exhaustion of compact cylinders $K_m \subset \mathbb{R}^3 \times \mathbb{R}$. For each k , choose $n(k)$ so large that $\lambda_{n(k), s_k} \rightarrow 0$ and the rescaled fields at scale $\lambda_{n(k)}$, shifted by s_k , are within $1/k$ of $(V(\cdot, \cdot + s_k), P(\cdot, \cdot + s_k) - c_k)$ on K_k in the local Seregin topology. This is possible because $\lambda_n \downarrow 0$ and because the original sequence converges locally to (V, P) .

By the assumed hull convergence, the shifted limit $(V(\cdot, \cdot + s_k), P(\cdot, \cdot + s_k) - c_k)$ converges to (W, Π) on each fixed compact cylinder. The diagonal sequence of physical rescalings with scales $\lambda_{n(k), s_k}$ therefore converges locally to (W, Π) . Hence (W, Π) is again a centered Type I Seregin limit at z_* . The assertion about nonvanishing follows from lower semicontinuity and local convergence of the CKN quantities after the prescribed pressure gauges. $\square$

5.3. Elementary consequences of stationary-family degeneracy. The stationary-family condition has two immediate PDE consequences. First, a differentiable nontrivial family produces a nontrivial element of the kernel of the linearized stationary Leray operator. Second, if an ancient trajectory is exactly constrained to a stationary family, then it is stationary.

Lemma 5.7 (Stationary family implies linearized degeneracy). *Let $a \mapsto W_a \in \mathfrak{B}$ be a C^1 stationary family satisfying (5.3). Then*

$$(5.8) \quad D\mathcal{F}(W_0)\dot{W}_0 = 0.$$

In particular, if $\dot{W}_0 \neq 0$ modulo infinitesimal rotations, the linearized stationary Leray operator has a genuine non-symmetry kernel element.

Proof. Differentiate $\mathcal{F}(W_a) = 0$ at $a = 0$. Since $\mathcal{F}$ is C^1 on $\mathfrak{B}$, this gives (5.8). The final statement is exactly the nontriviality condition in Theorem 5.2. $\square$

Lemma 5.8 (Trajectories constrained to a stationary family are stationary). *Let $a \mapsto W_a \in \mathfrak{B}$ be a C^1 stationary family with $\dot{W}_a \neq 0$ on an interval. Suppose that a smooth solution of (5.2) has the form*

$$(5.9) \quad V(\tau) = W_{a(\tau)}$$

for a C^1 function $a(\tau)$. Then a is constant, and hence V is stationary.

Proof. Since $\mathcal{F}(W_a) = 0$ for every a , equation (5.2) gives

$$a'(\tau)\dot{W}_{a(\tau)} = \partial_\tau V(\tau) = \mathcal{F}(W_{a(\tau)}) = 0.$$

Because $\dot{W}_{a(\tau)} \neq 0$, we obtain $a'(\tau) = 0$. Thus a is constant on the interval. $\square$

Corollary 5.9 (Exact stationary-family subcase). *If an element of the residual class is itself an exact trajectory on an admissible stationary family, then it is a stationary centered profile. Hence it is excluded whenever it satisfies the stationary L^3 Liouville criterion, or more generally whenever the stationary profile belongs to one of the previously closed stationary or structured classes.*

Proof. Let $V(\tau) = W_{a(\tau)}$ be such an exact stationary-family trajectory. By Theorem 5.8, $a'(\tau) = 0$ on every interval on which the stationary-family parametrization is nondegenerate. Hence a is constant and V is a single stationary centered profile. If that stationary profile satisfies the stationary L^3 Liouville criterion or lies in one of the previously closed structured or ordered lower residual classes, the corresponding prior exclusion removes it from the residual class. $\square$

5.4. Stationary-hull profiles which occur as Seregin limits. The remaining issue is not the internal dynamics along a stationary family; exact stationary-family motion is stationary by Theorem 5.8. The real issue is occurrence as a blow-up limit: can a degenerate stationary-family profile occur as a Seregin limit of a finite-energy suitable weak solution while retaining local concentration? The following statement is the precise Seregin limit occurrence statement required to close the degenerate stationary-hull class with retained local concentration. It is proved in the following section by the terminal local concentration theorem.

Theorem 5.10 (No degenerate stationary-family concentration profile occurs as a Seregin limit). *Fix an admissible stationary phase space $\mathfrak{B}$. Let (W, Π) be a stationary centered profile satisfying:*

- (i) $W \in \mathfrak{B}$, $\mathcal{F}(W) = 0$, and W lies on an admissible nontrivial stationary family;
- (ii) (W, Π) is a centered Type I Seregin limit of a finite-energy suitable weak solution at a singular point;
- (iii) (W, Π) is ε_* -concentrating;
- (iv) (W, Π) is not in the small, stationary L^3 , tight, closed structured/decay, axisymmetric bounded-circulation, or rotational relative-equilibrium classes already removed earlier in the stratification.

Then no such (W, Π) exists.

Proof. This is exactly Theorem 6.5, proved below from the local terminal stratification theorem. $\square$

Remark 5.11 (Interpretation of the Seregin limit occurrence theorem). Theorem 5.10 excludes occurrence as a Type I Seregin limit. It does not assert that degenerate stationary profiles cannot exist as elliptic solutions of the stationary centered equation. It asserts only that such profiles cannot occur as Type I blow-up limits with retained local concentration after all previously closed classes have been removed. This is the correct formulation for the residual problem: the elliptic family may exist, but it cannot be reached by the local blow-up procedure coming from a finite-energy suitable weak solution.

5.5. Closure of the degenerate stationary-hull concentration class.

Theorem 5.12 (Exclusion of the degenerate stationary-hull concentration class).

$$(5.10) \quad \mathcal{C}_{\text{stat}}^{\text{ret}}(\mathcal{S}; I, J; \mathfrak{B}) = \emptyset.$$

Consequently the ordered third residual class is empty, provided it is defined with the retained-concentration Seregin limit condition in [Theorem 5.4](#).

Proof. Suppose, for contradiction, that $V \in \mathcal{C}_{\text{stat}}^{\text{ret}}(\mathcal{S}; I, J; \mathfrak{B})$. By [Theorem 5.4](#), there is a stationary profile (W, Π) in the centered time-translation hull of (V, P) , and (W, Π) is ε_* -concentrating. The profile W belongs to $\mathfrak{B}$, satisfies $\mathcal{F}(W) = 0$, and lies on an admissible nontrivial stationary family. Moreover, by the ordered definition of $\mathcal{C}_{\text{stat}}^{\text{ret}}$, it is not in any of the previously closed lower classes.

By [Theorem 5.6](#), the hull profile (W, Π) is itself a centered Type I Seregin limit of the original finite-energy suitable weak solution at the same singular point, after reselecting scales. Since local concentration is retained, (W, Π) satisfies every condition in [Theorem 5.10](#). This contradicts the Seregin limit occurrence theorem. Therefore no such V exists, proving (5.10). $\square$

Proposition 5.13 (Already-closed integrable stationary-family subcase). *Let (W, Π) be a stationary centered profile lying on an admissible stationary family. If*

$$(5.11) \quad W \in L^3(\mathbb{R}^3),$$

then $W \equiv 0$. Hence no retained-concentration hull profile satisfying (5.11) can occur in $\mathcal{C}_{\text{stat}}^{\text{ret}}$.

Proof. The stationary physical representative associated with W is a backward self-similar Navier–Stokes solution. The assumption $W \in L^3(\mathbb{R}^3)$ places it in the stationary L^3 class already excluded by the classical Nečas–Růžička–Šverák Liouville theorem [11], equivalently by the endpoint ancient Liouville theorem recorded in Paper I and cited below as [Theorem 9.6](#). Hence $W \equiv 0$. A zero profile has zero pressure-normalized local CKN mass after choosing the admissible pressure gauge, so it cannot be ε_* -concentrating. $\square$

Remark 5.14 (Closure after terminal local concentration analysis). The preceding argument reduces the stationary-hull class with retained local concentration from a time-translation hull question to the question whether a stationary centered profile in a nontrivial stationary family can occur as a Seregin limit. [Theorem 5.10](#) is supplied by the terminal local concentration analysis in the next section.

Remark 5.15 (Recommended statement for the residual decomposition). For the residual decomposition, the third exclusion line should be stated as follows:

$$\begin{array}{c} \text{stationary-hull residual with retained concentration} \\ \xrightarrow{\text{stationary-family Seregin limit exclusion}} \emptyset. \end{array}$$

The precise class is the *degenerate stationary-hull residual class with retained local concentration*.

6. EXCLUSION OF DEGENERATE STATIONARY-HULL PROFILES WITH RETAINED CONCENTRATION

We close the remaining analytic statement in the degenerate stationary-hull class by the terminal local concentration theorem proved in the main paper. The argument is not specific to the existence of a non-isolated stationary family. Once a stationary-family element occurs as a Type I Seregin limit carrying retained local CKN concentration, it is a stationary terminal concentration profile. Such profiles are excluded by the same local terminal concentration analysis and separated-profile exclusion used for the final residual class. No global coercive estimate for the original solution is used.

Throughout this section, a centered Type I profile satisfies

$$(6.1) \quad \partial_\tau U - \Delta U + \frac{1}{2}y \cdot \nabla U + \frac{1}{2}U + (U \cdot \nabla)U + \nabla \Pi = 0, \quad \nabla \cdot U = 0$$

on $\mathbb{R}^3 \times \mathbb{R}$, with the usual local suitability and pressure-gauge conventions.

6.1. Terminal local concentration theorem. We use the following specialization of the terminal local concentration theorem proved below. It is stated here exactly in the form needed for the stationary-hull closure with retained local concentration.

Theorem 6.1 (Terminal local concentration exclusion theorem). *Let (U, Π) be a bounded centered Type I Seregin profile satisfying local suitability and carrying a retained local CKN concentration, i.e. for some compact parabolic cylinder Q and some $\varepsilon_* > 0$,*

$$(6.2) \quad \iint_Q (|U|^3 + |\Pi - c_Q(\tau)|^{3/2}) dy d\tau \geq \varepsilon_*.$$

Assume that (U, Π) is outside the previously closed small-amplitude, stationary L^3 , tight, closed structured/decay, axisymmetric bounded-circulation, and rotational relative-equilibrium classes. Then the following hold.

- (i) *Every terminal extraction of (U, Π) belongs to one of the local alternatives in [Theorem 8.6](#): a single retained concentration profile, local vanishing after removal of concentrating sequences, exterior escape of the concentrating sequence, diffuse exterior concentration, loss of local compactness, or a finite separated family of retained concentration profiles.*
- (ii) *Local vanishing, exterior escape, diffuse exterior concentration, and loss of local compactness cannot be the sole locations of the retained compact CKN concentration [\(6.2\)](#).*
- (iii) *Finite separated families of retained concentration profiles are impossible.*
- (iv) *Any remaining single retained concentration profile is terminally indecomposable: it has no retained concentrating tail limit, no diffuse exterior concentration, and no loss of local compactness along escaping recenterings.*
- (v) *Every retained terminally indecomposable concentration profile has a backward sequence of finite critical norm: there exist $\tau_k \rightarrow -\infty$ such that*

$$(6.3) \quad \sup_k \|U(\cdot, \tau_k)\|_{L^3(\mathbb{R}^3)} < \infty.$$

- (vi) *Retained terminal concentration profiles inherit mildness under the physical pullback. Therefore the Albritton–Barker endpoint Liouville theorem applies to any profile satisfying [\(6.3\)](#).*

Proof. This is [Theorem 9.14](#) applied to (U, Π) . The local alternatives in (i) are those of [Theorem 8.6](#). Item (ii) is the exclusion of local vanishing, exterior escape, diffuse exterior concentration, and local compactness failure in [Theorem 8.9](#). Item (iii) is [Theorem 9.13](#). Item (iv) is [Theorem 9.3](#). Item (v) is [Theorem 9.4](#), and item (vi) is [Theorems 9.5](#) and [9.6](#). These results are proved in the present paper from Paper 0, Paper I, and local concentration analysis; no global bound is assumed. $\square$

Remark 6.2 (Local nature of the theorem). The theorem above is a local concentration statement. It uses the local suitable-weak-solution compactness and pressure normalization developed from the Caffarelli–Kohn–Nirenberg theory [\[6\]](#), together with the Paper 0 local concentration input and the Paper I Seregin compactness theorem. It then uses terminal profile decomposition, exclusion of local vanishing, diffuse exterior concentration, and local compactness failure, and the finite separated-profile theorem. It does not assert or require a global a priori bound for the original physical solution.

For completeness, and to make the dependence on local estimates explicit, we record the elementary part of the argument which converts terminal indecomposability into sequence- L^3 control.

Lemma 6.3 (Indecomposable local profiles have backward sequence- L^3 control). *Let (U, Π) be a bounded smooth ancient solution of (6.1). Suppose that (U, Π) has retained local concentration and is terminally indecomposable: it has no retained concentrating tail limit, no diffuse exterior concentration, and no escaping recentering sequence along which local compactness, suitability, or pressure control fails. Then there exists a sequence $\tau_k \rightarrow -\infty$ such that*

$$(6.4) \quad \sup_k \|U(\cdot, \tau_k)\|_{L^3(\mathbb{R}^3)} < \infty.$$

Proof. We argue by contradiction. Assume that no sequence satisfying (6.4) exists. Then

$$(6.5) \quad \liminf_{\tau \rightarrow -\infty} \|U(\cdot, \tau)\|_{L^3(\mathbb{R}^3)} = \infty.$$

Choose $\tau_n \rightarrow -\infty$ such that $\int_{\mathbb{R}^3} |U(y, \tau_n)|^3 dy \rightarrow \infty$. Since U is bounded, we may choose radii $R_n \rightarrow \infty$ so that

$$(6.6) \quad \int_{|y| > R_n} |U(y, \tau_n)|^3 dy \geq 1.$$

Let

$$(6.7) \quad m_n := \sup_{|x| > R_n} \int_{B_1(x)} |U(y, \tau_n)|^3 dy.$$

If $\limsup_n m_n \geq \varepsilon_*$, then for a subsequence there exist $|x_n| > R_n$ such that

$$(6.8) \quad \int_{B_1(x_n)} |U(y, \tau_n)|^3 dy \geq \varepsilon_*.$$

After recentering at (x_n, τ_n) , either the local compactness and pressure control required for a terminal profile fail, or a subsequence converges in the terminal topology to a retained concentrating tail limit. Both alternatives contradict terminal indecomposability.

It remains to consider the case $m_n < \varepsilon_*$ for all sufficiently large n . Then (6.6) says that critical mass persists outside expanding balls, while (6.7) says that every unit-scale exterior recentering remains below the retained-concentration threshold. This is exactly diffuse exterior concentration, again contradicting terminal indecomposability. Thus (6.5) is impossible, and (6.4) follows. $\square$

6.2. Stationary concentration profiles are impossible.

Theorem 6.4 (No stationary residual concentration profile occurs as a Seregin limit). *Let (W, Π) be a stationary solution of the centered equation,*

$$(6.9) \quad -\Delta W + \frac{1}{2}y \cdot \nabla W + \frac{1}{2}W + (W \cdot \nabla)W + \nabla \Pi = 0, \quad \nabla \cdot W = 0.$$

Assume that:

- (i) (W, Π) is obtained as a centered Type I Seregin limit of a finite-energy suitable weak solution at a singular point;
- (ii) (W, Π) has retained local concentration in the sense of (6.2);
- (iii) (W, Π) is outside all previously closed lower classes listed in Theorem 6.1;

Then no such (W, Π) exists.

Proof. Suppose that such a stationary profile (W, Π) exists. Since it is a centered Type I Seregin limit and has retained local concentration, it is an admissible input for the terminal local concentration theorem. Apply Theorem 6.1 to (W, Π) .

Finite separated terminal concentration profiles are impossible by Theorem 6.1. Local vanishing, exterior escape, diffuse exterior concentration, and local compactness failure cannot be the only locations of the retained compact CKN concentration. Hence the retained concentration is

represented by a single terminal concentration profile. By the indecomposability conclusion in [Theorem 6.1](#), this single profile is terminally indecomposable.

Because (W, Π) is stationary in τ , the single terminal concentration profile may be represented by (W, Π) itself after the terminal normalization. Therefore [Theorem 6.3](#) gives a sequence $\tau_k \rightarrow -\infty$ such that

$$(6.10) \quad \sup_k \|W\|_{L^3(\mathbb{R}^3)} = \sup_k \|W(\cdot, \tau_k)\|_{L^3(\mathbb{R}^3)} < \infty.$$

Thus $W \in L^3(\mathbb{R}^3)$.

Let

$$(6.11) \quad w(x, t) = (-t)^{-1/2} W\left(\frac{x}{\sqrt{-t}}\right), \quad q(x, t) = (-t)^{-1} \Pi\left(\frac{x}{\sqrt{-t}}\right), \quad t < 0.$$

By [Theorem 9.5](#), (w, q) is a mild ancient Navier–Stokes solution after any finite time shift. The critical scaling gives

$$(6.12) \quad \|w(\cdot, t)\|_{L^3(\mathbb{R}^3)} = \|W\|_{L^3(\mathbb{R}^3)} < \infty \quad (t < 0).$$

Applying the Albritton–Barker endpoint Liouville theorem along any sequence $t_k \downarrow -\infty$, [Theorem 9.6](#), we obtain $w \equiv 0$, and hence $W \equiv 0$. This contradicts retained local concentration, since the pressure-normalized local CKN quantity of the zero profile is zero. Therefore no stationary residual concentration profile can occur as such a Seregin limit. $\square$

6.3. Closure of the degenerate stationary-hull residual class.

Corollary 6.5 (No degenerate stationary-family concentration profile occurs as a Seregin limit). *Fix an admissible stationary phase space $\mathfrak{B}$. There is no stationary centered profile (W, Π) satisfying all of the following conditions:*

- (i) $W \in \mathfrak{B}$, W solves the stationary centered equation (6.9), and W lies on an admissible nontrivial stationary family;
- (ii) (W, Π) is a centered Type I Seregin limit of a finite-energy suitable weak solution at a singular point;
- (iii) (W, Π) has retained local concentration;
- (iv) (W, Π) is outside the previously closed lower classes;

Proof. The stationary-family condition in (i) is additional structure, but the exclusion in [Theorem 6.4](#) applies to every stationary residual profile carrying retained local concentration. Hence it applies in particular to stationary profiles lying in admissible nontrivial stationary families. $\square$

Theorem 6.6 (Closure of the degenerate stationary-hull concentration class). *Let $\mathcal{C}_{\text{stat}}^{\text{ret}}(\mathcal{S}; I, J; \mathfrak{B})$ be the degenerate stationary-hull residual class with retained local concentration. Then*

$$(6.13) \quad \mathcal{C}_{\text{stat}}^{\text{ret}}(\mathcal{S}; I, J; \mathfrak{B}) = \emptyset.$$

Proof. Suppose, for contradiction, that $V \in \mathcal{C}_{\text{stat}}^{\text{ret}}(\mathcal{S}; I, J; \mathfrak{B})$. By the definition of the degenerate stationary-hull residual class with retained local concentration, the centered translation hull of (V, P) contains a stationary profile (W, Π) with retained local concentration which lies in an admissible nontrivial stationary family and is outside the previously closed lower classes. The scale-reselection lemma for centered hull limits shows that (W, Π) is itself a centered Type I Seregin limit of the original finite-energy suitable weak solution at the same singular point.

Thus (W, Π) satisfies the hypotheses of [Theorem 6.5](#). This is impossible. Hence no such V exists, and (6.13) follows. $\square$

Remark 6.7 (How this enters the residual decomposition). The third residual line should be recorded as a theorem closed by the terminal local concentration analysis:

$$\begin{aligned} & \text{degenerate stationary-hull residual with retained concentration} \\ & \implies \text{stationary profile with retained concentration} \implies \emptyset. \end{aligned}$$

The second implication is proved by terminal local concentration analysis: decomposable profiles produce either an excluded separated concentration family, diffuse exterior concentration, or loss of local compactness; indecomposable profiles have backward sequence- L^3 control and are ruled out by the endpoint ancient Liouville theorem.

7. TAIL GEOMETRY OF THE GENERIC TERMINAL RESIDUAL

We now pass to the generic terminal residual class. The preliminary point is that the centered equation is not invariant under raw fixed spatial translations. The correct far-field recenterings are self-similar covariant translations.

Definition 7.1 (Covariant translations). For $a \in \mathbb{R}^3$, define

$$(T_a V)(y, \tau) := V(y + e^{\tau/2} a, \tau), \quad (T_a P)(y, \tau) := P(y + e^{\tau/2} a, \tau).$$

Lemma 7.2 (Covariant translation invariance). *If (V, P) solves (1.2), then $(T_a V, T_a P)$ also solves (1.2).*

Proof. Set $z = y + e^{\tau/2} a$. Then

$$\partial_\tau (T_a V)(y, \tau) = (\partial_\tau V)(z, \tau) + \frac{1}{2} e^{\tau/2} a \cdot \nabla V(z, \tau).$$

Moreover,

$$\frac{1}{2} y \cdot \nabla (T_a V)(y, \tau) = \frac{1}{2} z \cdot \nabla V(z, \tau) - \frac{1}{2} e^{\tau/2} a \cdot \nabla V(z, \tau).$$

The two additional transport terms cancel in $\partial_\tau T_a V + \frac{1}{2} y \cdot \nabla T_a V$. The Laplacian, the zeroth-order term, the convection term, the pressure gradient, and the divergence constraint are translation covariant. Substitution into (1.2) proves the claim. $\square$

Remark 7.3 (Raw translates create a spurious drift). If one sets $V_n(y, \tau) = V(y + x_n, \tau)$ with fixed x_n , then the centered equation acquires the additional transport term

$$-\frac{1}{2} x_n \cdot \nabla V_n.$$

Thus fixed translates do not define a compact invariant far-field hull for the renormalized dynamics.

Definition 7.4 (Affine homogeneous symmetry). For $c \in \mathbb{R}^3$, define

$$(S_c V)(y, \tau) := V(y - 2c, \tau) - c, \quad (S_c P)(y, \tau) := P(y - 2c, \tau) + \frac{1}{2} c \cdot y.$$

Lemma 7.5 (Affine renormalized Galilean symmetry). *If (V, P) solves (1.2), then $(S_c V, S_c P)$ also solves (1.2).*

Proof. Set $U(y, \tau) = V(y - 2c, \tau) - c$, $\Pi(y, \tau) = P(y - 2c, \tau) + \frac{1}{2} c \cdot y$, and $z = y - 2c$. Then

$$\partial_\tau U = (\partial_\tau V)(z, \tau), \quad \Delta U = (\Delta V)(z, \tau), \quad \nabla U = (\nabla V)(z, \tau).$$

The nonlinear term is

$$(U \cdot \nabla) U = (V - c) \cdot \nabla V = (V \cdot \nabla) V - c \cdot \nabla V,$$

where all quantities on the right are evaluated at (z, τ) . The drift term satisfies

$$\frac{1}{2}y \cdot \nabla U = \frac{1}{2}(z + 2c) \cdot \nabla V = \frac{1}{2}z \cdot \nabla V + c \cdot \nabla V.$$

The two $c \cdot \nabla V$ terms cancel. Finally,

$$\frac{1}{2}U + \nabla \Pi = \frac{1}{2}(V - c) + \nabla P + \frac{1}{2}c = \frac{1}{2}V + \nabla P.$$

Substitution into (1.2) at the point (z, τ) gives the equation for (U, Π) . The divergence condition is unchanged by translation and subtracting a constant vector. $\square$

Definition 7.6 (Asymptotic hull at infinity). Let V be a bounded smooth ancient solution of (1.2). Define

$$\mathcal{A}_\infty(V) := \{W : \exists a_n \in \mathbb{R}^3, |a_n| \rightarrow \infty, T_{a_n}V \rightarrow W \text{ in } C_{\text{loc}}^\infty(\mathbb{R}^3 \times \mathbb{R})\}.$$

Theorem 7.7 (Compactness of the asymptotic hull). *Let V be a bounded smooth ancient solution of (1.2). Then $\mathcal{A}_\infty(V)$ is nonempty and compact in $C_{\text{loc}}^\infty(\mathbb{R}^3 \times \mathbb{R})$. Every $W \in \mathcal{A}_\infty(V)$ is a bounded smooth ancient solution of (1.2) with $\|W\|_{L^\infty} \leq \|V\|_{L^\infty}$.*

Proof. For any sequence a_n with $|a_n| \rightarrow \infty$, the translated solutions $T_{a_n}V$ solve the same equation and have the same L^∞ bound by Theorem 7.2. On every compact cylinder, the coefficients of (1.2) are fixed and smooth. Interior parabolic estimates for the Stokes system with bounded smooth nonlinear data, combined with the pressure equation $-\Delta P = \partial_i \partial_j (V_i V_j)$ after local gauges, give uniform C^k -bounds on smaller compact cylinders for every k . A diagonal Arzela–Ascoli argument gives a subsequential limit in C_{loc}^∞ . Passing to the limit in the equation gives another bounded ancient solution. The same compactness argument applied to an arbitrary sequence of elements of $\mathcal{A}_\infty(V)$ proves compactness. $\square$

Definition 7.8 (Tail alternatives). Let $\mathcal{C}$ denote the set of constant vector fields. Define

$$\mathcal{C}_{\text{gen tail-const}} := \{V \in \mathcal{C}_{\text{gen}} : \mathcal{A}_\infty(V) \subset \mathcal{C}\},$$

and

$$\mathcal{C}_{\text{gen tail-prof}} := \{V \in \mathcal{C}_{\text{gen}} : \mathcal{A}_\infty(V) \text{ contains a nonconstant profile}\}.$$

Proposition 7.9 (Tail dichotomy and tail-zero normalization). *Every $V \in \mathcal{C}_{\text{gen}}$ belongs to $\mathcal{C}_{\text{gen tail-const}} \cup \mathcal{C}_{\text{gen tail-prof}}$. If $\mathcal{A}_\infty(V) = \{C_\infty\}$ is a singleton constant hull, then*

$$U := S_{C_\infty}V$$

solves (1.2) and satisfies

$$\mathcal{A}_\infty(U) = \{0\}.$$

Proof. The dichotomy is the definition. Since T_a and S_c commute,

$$T_a(S_{C_\infty}V) = S_{C_\infty}(T_aV).$$

Thus every far-field limit of $S_{C_\infty}V$ is

$$S_{C_\infty}(C_\infty) = 0.$$

The equation is preserved by Theorem 7.5. $\square$

Definition 7.10 (Tail-minimal recurrent core). A nonempty compact set $\mathcal{M} \subset \mathcal{A}_\infty(V)$ is a tail-minimal recurrent core if it is invariant under all covariant translations T_a , invariant under all time translations

$$(\Theta_s W)(y, \tau) := W(y, \tau + s),$$

and contains no proper nonempty compact subset with the same two invariance properties.

Lemma 7.11 (Pressure gradients in invariant local averages). *Let $\mathcal{M}$ be a compact set of smooth bounded solutions of (1.2), compact in the locally smooth topology and invariant under the covariant translations T_a and the time translations Θ_s . Let μ be a Borel probability measure on $\mathcal{M}$ invariant under both actions. For each $W \in \mathcal{M}$, let P_W be any smooth local pressure near $(0, 0)$ satisfying (1.2). Put*

$$u_i(W) := W_i(0, 0), \quad p_i(W) := \partial_i P_W(0, 0), \quad b_{ij}(W) := u_i(W)u_j(W).$$

Let Π_0 denote the orthogonal projection in $L^2(\mathcal{M}, \mu)$ onto the closed subspace of functions invariant under all spatial covariant translations T_a , and set

$$u_i^0 := \Pi_0 u_i, \quad p_i^0 := \Pi_0 p_i.$$

Then the pressure contribution in the recurrent-core energy average is gauge-independent and satisfies

$$(7.1) \quad \int_{\mathcal{M}} u_i(W) p_i(W) d\mu(W) = \int_{\mathcal{M}} u_i^0(W) p_i^0(W) d\mu(W).$$

Equivalently, the nonzero spatial-frequency part of the pressure gradient does no work in invariant average:

$$(7.2) \quad \int_{\mathcal{M}} u_i(W) (p_i(W) - p_i^0(W)) d\mu(W) = 0.$$

Proof. The additive pressure gauge is irrelevant because only ∇P_W enters. The pressure gradient is fixed by the velocity through the equation

$$\nabla P_W = \Delta W - \frac{1}{2}(W + y \cdot \nabla W) - \partial_\tau W - (W \cdot \nabla)W.$$

Thus $W \mapsto p_i(W)$ is a continuous bounded local observable on $\mathcal{M}$. Paper II, [3, Lemma *Local pressure reconstruction*], supplies the local pressure representatives; the present argument uses only their gradients, so the additive gauges never enter.

Let U_a be the unitary representation of spatial covariant translations on $L^2(\mathcal{M}, \mu)$,

$$(U_a F)(W) := F(T_a W).$$

Let D_1, D_2, D_3 be its skew-adjoint infinitesimal generators. For a smooth local observable F , $D_k F$ is obtained by differentiating $F(T_{he_k} W)$ at $h = 0$. In particular, at $\tau = 0$,

$$D_k u_i(W) = \partial_k W_i(0, 0).$$

Hence

$$(7.3) \quad D_i u_i = 0 \quad \text{in } L^2(\mathcal{M}, \mu),$$

because each W is divergence-free.

Taking the spatial divergence of (1.2) gives the local pressure Poisson equation

$$(7.4) \quad -\Delta_y P_W = \partial_i \partial_j (W_i W_j).$$

Indeed, the divergence of $\partial_\tau W$, ΔW , and W is zero; the divergence of $y \cdot \nabla W$ is $\partial_i (y_j \partial_j W_i) = \partial_j W_j + y_j \partial_j \partial_i W_i = 0$; and $(W \cdot \nabla)W_i = \partial_j (W_j W_i)$. Applying the infinitesimal translation generators at $(0, 0)$ gives

$$(7.5) \quad D_i p_i = -D_i D_j b_{ij}.$$

The pressure gradient is curl-free, hence

$$(7.6) \quad D_k p_i = D_i p_k.$$

Let

$$A := - \sum_{k=1}^3 D_k^2.$$

By the spectral theorem for the commuting unitary group U_a , A is self-adjoint and nonnegative, and Π_0 is exactly the spectral projection of A at the eigenvalue 0. Define the nonzero spatial part

$$q_i := p_i - p_i^0 = (I - \Pi_0)p_i.$$

Equations (7.5) and (7.6) imply, on the nonzero spectral subspace,

$$q_k = D_k A^{-1} (I - \Pi_0) D_i D_j b_{ij}$$

in the standard regularized sense. Explicitly, with

$$q_{k,\varepsilon} := D_k (\varepsilon + A)^{-1} (I - \Pi_0) D_i D_j b_{ij}, \quad \varepsilon > 0,$$

the joint spectral resolution of D_1, D_2, D_3 gives

$$q_{k,\varepsilon} \longrightarrow q_k \quad \text{in } L^2(\mathcal{M}, \mu) \quad \text{as } \varepsilon \downarrow 0.$$

The convergence is just the multiplier convergence

$$\frac{\xi_k \xi_i \xi_j}{\varepsilon + |\xi|^2} \longrightarrow \frac{\xi_k \xi_i \xi_j}{|\xi|^2} \quad (\xi \neq 0),$$

together with $(I - \Pi_0)$, which removes the point $\xi = 0$. The curl-free relation (7.6) says that the nonzero spatial spectral part of p is parallel to ξ , and (7.5) fixes its longitudinal coefficient; there is therefore no additional harmonic component.

For each $\varepsilon > 0$, skew-adjointness of D_k and (7.3) give

$$\begin{aligned} \int_{\mathcal{M}} u_k q_{k,\varepsilon} d\mu &= \int_{\mathcal{M}} u_k D_k (\varepsilon + A)^{-1} (I - \Pi_0) D_i D_j b_{ij} d\mu \\ &= - \int_{\mathcal{M}} (D_k u_k) (\varepsilon + A)^{-1} (I - \Pi_0) D_i D_j b_{ij} d\mu = 0. \end{aligned}$$

Passing to the limit in L^2 yields

$$\int_{\mathcal{M}} u_k q_k d\mu = 0,$$

which is (7.2). Since p_i^0 lies in the range of Π_0 and $u_i - u_i^0$ is orthogonal to that range,

$$\int_{\mathcal{M}} (u_i - u_i^0) p_i^0 d\mu = 0.$$

Combining this with (7.2) proves (7.1). $\square$

Theorem 7.12 (Rigidity of recurrent profile tail cores). *Let $\mathcal{M}$ be a compact tail-minimal recurrent core for (1.2) arising from the Paper I normalized Seregine theorem, with the affine zero mode interpreted by the affine symmetry Theorem 7.5. Equivalently, after applying one fixed affine normalization S_c , the spatially homogeneous part of the core satisfies the bounded mild centered formulation of Paper II [3, Definition Bounded mild ancient solution]. Then every $W \in \mathcal{M}$ is spatially and temporally constant. More precisely, there exists $m \in \mathbb{R}^3$ such that*

$$W(y, \tau) \equiv m \quad \text{for every } W \in \mathcal{M}.$$

After applying the affine symmetry S_m , the core is reduced to the zero solution.

Proof. The group generated by covariant translations and time translations is amenable on the compact invariant set $\mathcal{M}$. Choose an invariant Borel probability measure μ with full support on a minimal component, and write

$$\langle F \rangle := \int_{\mathcal{M}} F(W) d\mu(W)$$

for local observables. Let P_W be a local pressure associated to W . As in [Theorem 7.11](#), set

$$u_i(W) := W_i(0, 0), \quad p_i(W) := \partial_i P_W(0, 0),$$

and let Π_0 be the orthogonal projection onto observables invariant under all spatial covariant translations. Write

$$u_i^0 := \Pi_0 u_i, \quad p_i^0 := \Pi_0 p_i.$$

Invariance gives

$$\langle \partial_\tau F(W)(0, 0) \rangle = 0, \quad \langle \partial_j F(W)(0, 0) \rangle = 0$$

for every smooth local observable F . These identities follow first for difference quotients generated by the group action and then by smoothness.

Evaluate [\(1.2\)](#) at $(0, 0)$. The drift term vanishes at $y = 0$, and we get

$$\partial_\tau W(0, 0) - \Delta W(0, 0) + \frac{1}{2} W(0, 0) + (W(0, 0) \cdot \nabla) W(0, 0) + \nabla P_W(0, 0) = 0.$$

Project this identity onto the spatially invariant subspace. The spatial derivative terms vanish under Π_0 : the Laplacian is $D_j D_j u_i$, and the nonlinear term is $D_j(u_j u_i)$ because $D_j u_j = 0$. Since time translations commute with spatial translations, $\Pi_0 \partial_\tau u_i = \partial_\tau u_i^0$. Thus

$$(7.7) \quad \partial_\tau u_i^0 + \frac{1}{2} u_i^0 + p_i^0 = 0 \quad \text{in } L^2(\mathcal{M}, \mu).$$

Taking the inner product of [\(7.7\)](#) with u_i^0 , summing in i , and using invariance under time translations,

$$\langle u_i^0 \partial_\tau u_i^0 \rangle = \frac{1}{2} \langle \partial_\tau |u^0|^2 \rangle = 0,$$

gives

$$(7.8) \quad \langle u_i^0 p_i^0 \rangle = -\frac{1}{2} \langle |u^0|^2 \rangle.$$

Dot the equation with W , evaluate at $(0, 0)$, and average. The time term is

$$\langle W \cdot \partial_\tau W \rangle = \frac{1}{2} \langle \partial_\tau |W|^2 \rangle = 0.$$

The diffusion term gives

$$-W \cdot \Delta W = |\nabla W|^2 - \operatorname{div}(W \cdot \nabla W),$$

and therefore

$$-\langle W \cdot \Delta W \rangle = \langle |\nabla W|^2 \rangle.$$

The nonlinear term is a divergence:

$$W \cdot (W \cdot \nabla) W = \frac{1}{2} W \cdot \nabla |W|^2 = \frac{1}{2} \operatorname{div}(|W|^2 W),$$

so its average is zero. Decompose

$$p_i = p_i^0 + (p_i - p_i^0).$$

The nonzero spatial part of the pressure gradient does no work in invariant average by [Theorem 7.11](#), and the remaining spatial zero mode is evaluated by [\(7.8\)](#). Hence

$$\langle u_i p_i \rangle = \langle u_i^0 p_i^0 \rangle = -\frac{1}{2} \langle |u^0|^2 \rangle.$$

Thus

$$(7.9) \quad \langle |\nabla W|^2 \rangle + \frac{1}{2} \langle |W|^2 \rangle - \frac{1}{2} \langle |u^0|^2 \rangle = 0.$$

Both terms are nonnegative. The second nonnegative quantity is

$$\langle |u|^2 \rangle - \langle |u^0|^2 \rangle = \langle |u - u^0|^2 \rangle,$$

because Π_0 is an orthogonal projection in $L^2(\mathcal{M}, \mu)$. Therefore

$$\langle |\nabla W|^2 \rangle = 0, \quad \langle |u - u^0|^2 \rangle = 0.$$

Since μ has full support and the observables are continuous in the local smooth topology, $\nabla W(0, 0) = 0$ for every $W \in \mathcal{M}$. Invariance moves $(0, 0)$ to any (y, τ) , so $\nabla W(y, \tau) = 0$ everywhere. Thus every element of $\mathcal{M}$ is spatially constant:

$$W(y, \tau) = a_W(\tau).$$

It remains only to identify the allowed spatially homogeneous zero mode. This is not a pressure-gauge assertion. By the Paper I normalization quoted in the statement, there is one fixed affine normalization S_c for which the spatially homogeneous representatives satisfy the bounded mild centered formulation of Paper II. For a spatially constant mild centered solution $B(\tau)$, the nonlinear Duhamel term in Paper II's mild formula vanishes and the Ornstein–Uhlenbeck–Stokes semigroup of Paper II acts on constants by

$$S(\tau - \sigma)B(\sigma) = e^{-(\tau - \sigma)/2} B(\sigma).$$

Thus

$$B(\tau) = e^{-(\tau - \sigma)/2} B(\sigma) \quad (\sigma < \tau).$$

Letting $\sigma \rightarrow -\infty$ and using boundedness of the ancient trajectory gives $B(\tau) = 0$ for every τ . Undoing the fixed affine normalization gives a single vector $m = c$ and

$$W(y, \tau) \equiv m \quad (W \in \mathcal{M}).$$

The affine symmetry S_m then maps the core to the zero solution. $\square$

8. TERMINAL CONCENTRATION PROFILES AND SEPARATED CENTERS

The terminal part of the proof is local in parabolic cylinders. It does not require a global L^3 bound, a global CKN budget, a finite critical-mass budget, or a global nonlinear profile decomposition for the original residual profile. Everything is formulated on fixed compact terminal cylinders after applying the recentering sequence under consideration.

Definition 8.1 (Parabolic recentering sequence). Let (V_n, P_n) be a terminal extraction sequence on expanding domains $D_n \uparrow \mathbb{R}^3 \times \mathbb{R}$, with local suitability and pressure-gauge control. A parabolic recentering sequence is a sequence

$$\mathbf{c} = (z_n)_{n \geq 1}, \quad z_n = (y_n, \tau_n) \in D_n.$$

Two recentering sequences are equivalent if

$$\sup_n \left(|y_n - y'_n| + |\tau_n - \tau'_n|^{1/2} \right) < \infty,$$

and asymptotically separated if the same parabolic distance tends to infinity. The recentered fields associated to $\mathfrak{c}$ are denoted

$$V_n^\mathfrak{c}(y, s) := V_n(y + y_n, s + \tau_n), \quad P_n^\mathfrak{c}(y, s) := P_n(y + y_n, s + \tau_n) - a_n^\mathfrak{c}(s),$$

where $a_n^\mathfrak{c}$ is an admissible time-dependent local pressure gauge. All convergence assertions for a recentering sequence are made on fixed compact sets in the variables (y, s) .

Definition 8.2 (Local CKN measure and concentrating recentering sequences). After choosing local pressure gauges $a_n(\tau)$, define

$$d\mu_n := \left(|V_n|^3 + |P_n - a_n(\tau)|^{3/2} \right) dy d\tau, \quad d\nu_n := |\nabla V_n|^2 dy d\tau.$$

The local mass of a recentering sequence $\mathfrak{c} = (z_n)$ is

$$m(\mathfrak{c}) := \limsup_{n \rightarrow \infty} \mu_n(Q_1(z_n)).$$

For a fixed small-data threshold $\varepsilon_{\text{sd}} > 0$, the recentering sequence is ε_{sd} -concentrating if $m(\mathfrak{c}) \geq \varepsilon_{\text{sd}}$, and locally vanishing at unit scale otherwise. It has loss of local compactness if

$$\limsup_{n \rightarrow \infty} \nu_n(Q_2(z_n)) = \infty$$

or if the local pressure gauges fail to be uniformly controlled in $L^{3/2}(Q_2(z_n))$.

Lemma 8.3 (Paper I compactness excludes retained local compactness failure). *Let (V_n, P_n) be a terminal sequence obtained from a Paper I Type I Seregin extraction, or from a later terminal extraction of such a sequence. If a recentering sequence $\mathfrak{c} = (z_n)$ carries retained compact velocity L^3 -mass, then after passing to a subsequence it has no loss of local compactness and its recentered fields converge locally in the Seregin topology to a suitable bounded terminal profile.*

Proof. Fix $R < \infty$. The recentered cylinders $Q_{2R}(z_n)$ are compact terminal cylinders in the variables of the chosen extraction. Paper I gives local suitability, local pressure-gauge compactness, and the inherited Type I bound on every such compact cylinder. Therefore, after choosing the pressure gauges $a_n^\mathfrak{c}$, the sequence $(V_n^\mathfrak{c}, P_n^\mathfrak{c})$ has the standard local bounds needed for the Seregin compactness theorem on $Q_R(0, 0)$: strong local L^3 compactness for the velocity, weak local $L^{3/2}$ compactness for the pressure, and the local energy inequality in the limit. A diagonal argument over $R = 1, 2, \dots$ gives a suitable bounded terminal profile. The two failures used in the definition of loss of local compactness are precisely the negations of these local compactness and pressure-gauge conclusions, so a retained recentering sequence arising from the Paper I Type I extraction theorem cannot remain in the local-compactness-failure alternative. $\square$

Lemma 8.4 (Maximal separated families of concentrating recenterings). *There exists a maximal separated family of ε_{sd} -concentrating recentering sequences without loss of local compactness. After this family is extracted, every residual unit parabolic cylinder outside sufficiently large tubes around the selected recentering sequences has CKN mass below ε_{sd} .*

Proof. The existence of a maximal separated concentrating family follows by Zorn's lemma on the partially ordered set of separated concentrating families. If residual local vanishing failed after extraction, there would be centers z'_n outside all large selected tubes such that

$$\limsup_{n \rightarrow \infty} \mu_n(Q_1(z'_n)) \geq \varepsilon_{\text{sd}}.$$

Such a recentering sequence is ε_{sd} -concentrating and asymptotically separated from every selected concentrating recentering sequence, contradicting maximality. Hence the residual after maximal concentration extraction is locally vanishing at unit scale. $\square$

Lemma 8.5 (Separation of recentered cylinders). *Let $\mathbf{c}_i = (z_{i,n})$ and $\mathbf{c}_j = (z_{j,n})$ be asymptotically separated recentering sequences. For every fixed $R < \infty$, the cylinder $Q_R(z_{j,n})$, written in the coordinates of $\mathbf{c}_i$, leaves every compact subset of the recentered variables. Consequently no positive compact velocity L^3 -mass assigned to $\mathbf{c}_j$ is visible on any fixed compact cylinder of the $\mathbf{c}_i$ -frame.*

Proof. In the $\mathbf{c}_i$ -coordinates the center of $Q_R(z_{j,n})$ is

$$(y_{j,n} - y_{i,n}, \tau_{j,n} - \tau_{i,n}).$$

The parabolic distance of this point from the origin tends to infinity by asymptotic separation. Hence, for every fixed compact terminal cylinder K , one has

$$K \cap (Q_R(z_{j,n}) - z_{i,n}) = \emptyset$$

for all sufficiently large n . This is a purely local statement about the relative positions of the two recentering sequences. It does not estimate the solution on all of space and does not use a global norm. $\square$

Theorem 8.6 (Terminal local concentration alternatives). *Every terminal concentration sequence belongs to exactly one of the following alternatives:*

$$\mathcal{S}_{\text{one}}, \quad \mathcal{S}_{\text{locvan}}, \quad \mathcal{S}_{\text{ext}}, \quad \mathcal{S}_{\text{diff}}, \quad \mathcal{S}_{\text{noncomp}}, \quad \mathcal{S}_{\text{sep}}.$$

Here $\mathcal{S}_{\text{noncomp}}$ consists of recentering sequences with loss of local compactness; $\mathcal{S}_{\text{one}}$ and $\mathcal{S}_{\text{sep}}$ are the bounded concentrating alternatives with respectively one, or at least two, bounded separated equivalence classes; $\mathcal{S}_{\text{ext}}$ consists of concentrating recentering sequences escaping the terminal frame; $\mathcal{S}_{\text{diff}}$ consists of diffuse exterior concentration, namely critical mass persisting on expanding regions while every fixed unit cylinder is locally vanishing; and $\mathcal{S}_{\text{locvan}}$ is local CKN vanishing after removal of the selected concentrating recentering sequences.

Proof. Let $\mathbf{c}$ be a terminal concentration sequence. If it has loss of local compactness, it belongs to $\mathcal{S}_{\text{noncomp}}$. Assume it has no loss of local compactness. If it is ε_{sd} -concentrating, maximal concentration extraction puts it into the selected concentrating family up to equivalence. If its representative remains bounded in the terminal frame and it is the unique bounded concentrating class, it gives the single-profile alternative. If more than one bounded concentrating class is present, it gives a finite separated profile family. If it escapes every bounded terminal frame, it is the exterior escape alternative. This classifies concentrating sequences without loss of local compactness.

It remains to classify residual sequences which are locally vanishing at unit scale and have no loss of local compactness. By Theorem 8.4, the residual after concentration extraction has local CKN vanishing on every fixed compact terminal cylinder. If no critical mass remains on expanding regions, this is the local-vanishing alternative $\mathcal{S}_{\text{locvan}}$. If critical mass persists only on expanding regions while every fixed unit cylinder remains below ε_{sd} , this is the diffuse exterior concentration alternative $\mathcal{S}_{\text{diff}}$. This is a terminal alternative, not an a priori global estimate on the original solution. The alternatives are mutually exclusive by construction and exhaustive by the preceding cases. $\square$

Corollary 8.7 (Persistent local velocity mass forces retained concentration). *If a terminal sequence satisfies*

$$\liminf_{n \rightarrow \infty} \int \int_{Q_1(0,0)} |V_n|^3 dy d\tau \geq \eta_0 > 0$$

with $\eta_0 \geq \varepsilon_{\text{sd}}$, then the terminal decomposition contains a local concentration profile unless the distinguished recentering sequence has loss of local compactness. In the generic residual after local vanishing and local compactness failure are excluded, the persistent compact velocity mass

is therefore carried by either one retained concentration profile or a finite separated family of retained concentration profiles.

Proof. The local CKN measure μ_n dominates the velocity contribution. Hence the origin recentering sequence has μ -mass at least η_0 , and is ε_{sd} -concentrating unless it has loss of local compactness. If it has no loss of local compactness and is bounded in the terminal frame, it lies in the single-profile or separated-profile alternative of [Theorem 8.6](#). $\square$

Lemma 8.8 (Local vanishing and exterior alternatives do not carry compact concentration). *Let a terminal extraction inherit the positive compact velocity L^3 -mass supplied by Paper 0 and Paper I. Then the retained compact velocity mass cannot be carried solely by local vanishing, exterior escape, or diffuse exterior concentration.*

Proof. The assertion is local. By Paper 0, failure of local vanishing at a singular terminal point gives a positive local velocity concentration sequence. Paper I carries this through the Type I Seregin extraction as the compact velocity lower bound (1.5). Cover the fixed compact cylinder K in (1.5) by finitely many unit terminal cylinders. At least one cylinder in the cover carries a positive velocity lower bound, say $\eta_K > 0$. Since μ_n dominates $|V_n|^3 dy d\tau$, choosing $\varepsilon_{\text{sd}} \leq \eta_K$ makes that unit cylinder ε_{sd} -concentrating. Thus the distinguished compact concentration cannot be locally vanishing.

Local vanishing after removal of concentrating recentering sequences means that the residual state has local CKN mass below ε_{sd} on each fixed compact terminal cylinder; it therefore cannot contain the fixed compact velocity lower bound. An exterior escaping concentration is visible only after its own recentering; relative to the distinguished compact terminal cylinder it escapes every bounded window. By [Theorem 8.5](#), it cannot be the sole location of a lower bound that remains in the distinguished compact window. Finally, diffuse exterior concentration is defined by persistence through expanding regions while all fixed unit cylinders are locally vanishing. If such a sequence contained the compact Paper 0/Paper I velocity lower bound, the positive concentration lemma of Paper 0 would select a unit terminal cylinder with mass at least the retained threshold, contradicting local vanishing. Thus none of these alternatives can be the unique location of the retained compact velocity mass. $\square$

Theorem 8.9 (Local exclusion of non-compact and non-concentrating terminal alternatives). *For terminal sequences generated by the Paper I Type I Seregin extraction theorem, loss of local compactness, local vanishing, exterior escape, and diffuse exterior concentration are not possible sole locations of the retained compact velocity mass.*

Proof. Retained recentering sequences with loss of local compactness are removed by [Theorem 8.3](#). Local vanishing, exterior escape, and diffuse exterior concentration are removed by [Theorem 8.8](#). The proof uses only the local positive concentration theorem of Paper 0, the local Type I Seregin compactness and pressure gauges of Paper I, and fixed compact terminal cylinders. $\square$

Definition 8.10 (Finite separated family of retained concentration profiles). A finite separated family of retained concentration profiles is a finite family of parabolic recentering sequences $g_{1,n}, \dots, g_{J,n}$, $J \geq 2$, such that:

- (i) the recentering sequences are mutually separated;
- (ii) each recentered sequence converges locally smoothly for velocity and locally in $L^{3/2}$ for pressure modulo gauges to a terminal profile (U_j, Π_j) ;
- (iii) each limiting profile has retained local concentration:

$$\mathcal{E}_1(U_j, \Pi_j; 0) \geq \varepsilon_* \quad (j = 1, \dots, J).$$

This is a finite local family, not a statement that all concentration profiles in the whole space have been counted. The separated-profile proof below treats the possible continuation of escaping recenterings by local concentration alternatives, not by a global counting estimate.

Proposition 8.11 (Finite separated families reduce to single-profile alternatives). *Assume the single-profile retained concentration closure statement: no single retained concentration profile remains in the refined residual decomposition after the already closed local classes are removed. Then a finite separated family is not a new refined residual alternative.*

Proof. Let $A = \{\mathfrak{s}_1, \dots, \mathfrak{s}_J\}$ be a finite separated concentrating family, represented by mutually separated parabolic recentering sequences. Recenter the terminal sequence at the sequence defining $\mathfrak{s}_j$. In that frame all other concentration profiles leave every compact terminal cylinder, so the local terminal limit is a single retained concentration profile. By the single-profile closure statement, this component either belongs to the already closed local classes, has local vanishing, or has loss of local compactness. Since the original family has retained concentration and local compactness, the latter two alternatives are incompatible with the hypotheses. Applying this argument to each j shows that the finite separated family consists only of already closed local profiles, and hence does not remain in the refined residual decomposition. $\square$

9. SEPARATED CONCENTRATION PROFILES, INDECOMPOSABILITY, AND SEQUENCE- L^3

We now close the terminal alternatives carrying retained local concentration. The proof uses no global critical-mass budget. Instead, finite separated concentration families are treated through local recentering and compactness. Iterating retained tail limits from such a finite family has only three local outcomes: an indecomposable retained concentration profile, a finite recurrent tail loop, or escape into one of the non-concentrating or non-compact alternatives already excluded by [Theorem 8.9](#).

Definition 9.1 (Tail limits and terminal indecomposability). Let (U, Π) be a terminal profile with retained local concentration. A retained concentrating tail limit is a terminal limit obtained from recentering sequences escaping to infinity relative to U , with retained local concentration at the origin. Diffuse exterior concentration means that there are $\tau_n \rightarrow -\infty$, $R_n \rightarrow \infty$, and $\eta > 0$, such that

$$\int_{|y| > R_n} |U(y, \tau_n)|^3 dy \geq \eta,$$

while every unit-scale exterior recentering sequence remains below the retained-concentration threshold, unless that recentering sequence has loss of local compactness. An escaping non-compact recentering sequence is a tail recentering sequence for which the local compactness, suitability, or pressure control needed to extract a terminal profile fails. A retained concentration profile is terminally indecomposable if it has no retained concentrating tail limit, no diffuse exterior concentration, and no escaping non-compact recentering sequence.

Lemma 9.2 (One retained tail limit gives two separated concentration profiles). *Let U be a retained concentration terminal profile. If U has a retained concentrating tail limit, then the corresponding terminal extraction contains a finite separated family of retained concentration profiles.*

Proof. The origin recentering sequence carries retained concentration:

$$\mathcal{E}_1(U, \Pi; 0) \geq \varepsilon_*.$$

Let W be a retained concentrating tail limit. By definition there are relative tail recentering sequences $h_n \rightarrow \infty$ such that $h_n \cdot U \rightarrow W$ in the terminal topology and

$$\mathcal{E}_1(W, \Pi_W; 0) \geq \varepsilon_*.$$

By local convergence of velocity and pressure after gauges,

$$\mathcal{E}_1(h_n \cdot U, \Pi_{h_n}; 0) \geq \frac{1}{2} \varepsilon_*$$

for all sufficiently large n . Thus, for large n , the origin recentering sequence and the sequence h_n are separated and have retained local concentration. They form a finite separated two-profile concentration family. $\square$

Proposition 9.3 (No finite separated family and local exclusions imply indecomposability). *Every single retained concentration terminal profile arising from a refined residual candidate is terminally indecomposable.*

Proof. Let U be a single retained concentration terminal profile. If U had a retained concentrating tail limit, [Theorem 9.2](#) would produce a finite two-profile separated family with retained concentration, contradicting [Theorem 9.13](#). If U had diffuse exterior concentration or an escaping non-compact recentering sequence, the corresponding terminal extraction would belong to one of the terminal alternatives excluded by [Theorem 8.9](#). Hence no retained concentrating tail limit, diffuse exterior concentration, or escaping non-compact recentering sequence exists, and U is terminally indecomposable. $\square$

Lemma 9.4 (Indecomposable sequence- L^3 completeness). *Let (U, Π) be a bounded smooth ancient solution of (1.2). Suppose U has retained local concentration and is terminally indecomposable in the sense of [Theorem 9.1](#). Then there exists a sequence $\tau_k \rightarrow -\infty$ such that*

$$\sup_k \|U(\cdot, \tau_k)\|_{L^3(\mathbb{R}^3)} < \infty.$$

Proof. We prove the contrapositive. Define

$$A(\tau) := \int_{\mathbb{R}^3} |U(y, \tau)|^3 dy \in [0, \infty].$$

If the conclusion fails, then

$$\liminf_{\tau \rightarrow -\infty} A(\tau) = \infty.$$

Choose $\tau_n \rightarrow -\infty$ such that $A(\tau_n) \rightarrow \infty$, allowing the value $+\infty$. Since U is bounded by some M , for each finite R ,

$$\int_{|y| \leq R} |U(y, \tau_n)|^3 dy \leq M^3 |B_R|.$$

Choose $R_n \rightarrow \infty$ so slowly that, after discarding finitely many n ,

$$\int_{|y| > R_n} |U(y, \tau_n)|^3 dy \geq 1.$$

If $A(\tau_n) = +\infty$, take $R_n = n$; otherwise choose R_n with $M^3 |B_{R_n}| \leq A(\tau_n)/2$.

Set

$$m_n := \sup_{|x| > R_n} \int_{B_1(x)} |U(y, \tau_n)|^3 dy.$$

There are two alternatives. First suppose $\limsup_{n \rightarrow \infty} m_n \geq \varepsilon_*$. Then, passing to a subsequence, there are x_n with $|x_n| > R_n \rightarrow \infty$ and

$$\int_{B_1(x_n)} |U(y, \tau_n)|^3 dy \geq \varepsilon_*.$$

Use the covariant tail recentering

$$U_n(y, s) := U(y + e^{s/2} x_n, \tau_n + s).$$

At $s = 0$,

$$\int_{B_1(0)} |U_n(y, 0)|^3 dy \geq \varepsilon_*.$$

If this recentering sequence has loss of local compactness, this contradicts terminal indecomposability. If it has local compactness, the compactness theorem gives a subsequential smooth limit W , and

$$\int_{B_1(0)} |W(y, 0)|^3 dy \geq \varepsilon_*,$$

so W is a retained concentrating tail limit, again contradicting terminal indecomposability.

The remaining alternative is that $m_n < \varepsilon_*$ for all sufficiently large n . Together with the exterior lower bound, this says that at times τ_n there is at least one unit of L^3 -mass outside B_{R_n} , but no unit-scale exterior recentering sequence carries retained local concentration. If some such exterior recentering sequence has loss of local compactness, terminal indecomposability is contradicted. If no such recentering sequence has loss of local compactness, this is diffuse exterior concentration with $\eta = 1$, again contradicting terminal indecomposability. Both alternatives are impossible, so the desired sequence exists. $\square$

Lemma 9.5 (Mildness inheritance for terminal profiles). *If U is a terminal profile with retained local concentration, then the physical pullback*

$$u(x, t) = (-t)^{-1/2} U\left(\frac{x}{\sqrt{-t}}, -\log(-t)\right), \quad t < 0,$$

is a mild ancient Navier–Stokes solution after any finite time shift.

Proof. Fix $T < 0$. On $\mathbb{R}^3 \times (-\infty, T]$, the pullback is smooth and bounded by $(-T)^{-1/2} \|U\|_{L^\infty}$. It solves the physical Navier–Stokes equations by the direct self-similar change of variables. For $-\infty < s < t \leq T$, applying the Duhamel formula to the smooth bounded solution on $[s, t]$, with the Leray projection interpreted in the same distributional-duality sense as the bounded mild formulation of Paper II [3], gives

$$u(t) = e^{(t-s)\Delta} u(s) - \int_s^t e^{(t-\sigma)\Delta} \mathbb{P} \nabla \cdot (u \otimes u)(\sigma) d\sigma.$$

After translating the terminal time T to 0, this is exactly the mild ancient formulation on every finite forward interval. Since $T < 0$ was arbitrary, the pullback is mild after any finite time shift. $\square$

Theorem 9.6 (Albritton–Barker endpoint Liouville theorem). *Let u be a mild ancient solution of the three-dimensional Navier–Stokes equations. If there exists $t_k \downarrow -\infty$ such that*

$$\sup_k \|u(\cdot, t_k)\|_{L^3(\mathbb{R}^3)} < \infty,$$

then $u \equiv 0$.

Proof. This is the endpoint ancient Liouville theorem in the sequence- L^3 form proved by Albritton and Barker [5]. Paper I records the same peer-reviewed input among its auxiliary Liouville results [2]. $\square$

Lemma 9.7 (No retained indecomposable concentration profile). *No retained terminally indecomposable concentration profile exists.*

Proof. Let U have retained local concentration and be terminally indecomposable. By Theorem 9.4, choose $\tau_k \rightarrow -\infty$ such that

$$\sup_k \|U(\cdot, \tau_k)\|_{L^3} < \infty.$$

Let u be the physical pullback and set $t_k = -e^{-\tau_k}$. Since $\tau_k \rightarrow -\infty$, the sequence satisfies $t_k \downarrow -\infty$, and critical scaling gives

$$\|u(\cdot, t_k)\|_{L^3(\mathbb{R}^3)} = \|U(\cdot, \tau_k)\|_{L^3(\mathbb{R}^3)}.$$

By [Theorem 9.5](#), u is mild ancient; hence [Theorem 9.6](#) gives $u \equiv 0$, and therefore $U \equiv 0$. This contradicts retained local concentration. $\square$

Lemma 9.8 (Diagonal lifting of retained concentrating tail limits). *Let $A = \{\mathfrak{s}_1, \dots, \mathfrak{s}_J\}$, $J \geq 2$, be a finite retained concentration separated family for a refined residual candidate, represented by separated recentering sequences $g_{j,n} = (y_{j,n}, \tau_{j,n})$. Let (U_j, Π_j) be the terminal profile obtained in the j -th recentering. If (U_j, Π_j) has a retained concentrating tail limit (W, Π_W) , then there is a parabolic recentering sequence $\mathfrak{d} = (d_n)$ for the original terminal sequence such that:*

- (i) $\mathfrak{d}$ is asymptotically separated from g_j ;
- (ii) the recentered original sequence at $\mathfrak{d}$ converges locally to (W, Π_W) , after admissible pressure gauges;
- (iii) $\mathfrak{d}$ has retained local concentration.

Proof. By definition of retained concentrating tail limit, there are relative tail recenterings $\zeta_m = (\xi_m, \sigma_m)$, with

$$|\xi_m| + |\sigma_m|^{1/2} \rightarrow \infty,$$

such that the fields obtained from (U_j, Π_j) by recentering at ζ_m converge locally in the terminal topology to (W, Π_W) , and

$$\mathcal{E}_1(W, \Pi_W; 0) \geq \varepsilon_*.$$

Fix m . Since $V_n^{g_j} \rightarrow U_j$ locally smoothly and $(P_n^{g_j}) \rightarrow \Pi_j$ locally in $L^{3/2}$ modulo gauges, the convergence also holds on the fixed compact cylinder $Q_m(\zeta_m)$. Therefore one can choose $n(m)$ so large that the recentered original fields

$$V_{n(m)}(\cdot + y_{j,n(m)} + \xi_m, \cdot + \tau_{j,n(m)} + \sigma_m)$$

and the corresponding pressure gauges are within m^{-1} of the ζ_m -recentered fields of (U_j, Π_j) on $Q_m(0, 0)$.

Set

$$d_m := (y_{j,n(m)} + \xi_m, \tau_{j,n(m)} + \sigma_m).$$

Relabeling the subsequence $n(m)$ as the terminal index gives a recentering sequence $\mathfrak{d} = (d_m)$ for the original terminal sequence. Since parabolic distance from d_m to $g_{j,n(m)}$ is exactly $|\xi_m| + |\sigma_m|^{1/2}$, the sequence $\mathfrak{d}$ is separated from the j -th recentering sequence. The diagonal choice gives local convergence of the original sequence at $\mathfrak{d}$ to (W, Π_W) . Finally, the local CKN quantity is continuous under the strong local velocity convergence and weak lower semicontinuity of the pressure term after the fixed gauges, so

$$\liminf_{m \rightarrow \infty} \mu_{n(m)}(Q_1(d_m)) \geq \mathcal{E}_1(W, \Pi_W; 0) \geq \varepsilon_*.$$

Thus $\mathfrak{d}$ has retained local concentration. $\square$

Lemma 9.9 (Escaping recenterings in a separated family). *Let $A = \{\mathfrak{s}_1, \dots, \mathfrak{s}_J\}$, $J \geq 2$, be a finite retained concentration separated family for a refined residual candidate. For each component $\mathfrak{s}_j$, exactly one of the following alternatives holds:*

- (i) the profile (U_j, Π_j) is terminally indecomposable;
- (ii) (U_j, Π_j) has diffuse exterior concentration or an escaping recentering sequence with loss of local compactness;
- (iii) (U_j, Π_j) has a retained concentrating tail limit which lifts to a recentering sequence of the original terminal sequence with retained local concentration.

Proof. This is the definition of terminal indecomposability plus the diagonal lifting lemma. If (U_j, Π_j) is not terminally indecomposable, then at least one of the three excluded alternatives in [Theorem 9.1](#) occurs: a retained concentrating tail limit, diffuse exterior concentration, or an escaping recentering sequence with loss of local compactness. The diffuse exterior and local-compactness-failure cases give (ii). In the retained concentrating tail-limit case, [Theorem 9.8](#) lifts the limit to a recentering sequence of the original terminal sequence with retained local concentration, giving (iii). The three alternatives are disjoint by definition. $\square$

Lemma 9.10 (No diffuse or non-compact escaping recentering from a retained family). *In the setting of [Theorem 9.9](#), alternative (ii) cannot occur.*

Proof. Suppose that (U_j, Π_j) has diffuse exterior concentration or an escaping recentering sequence with loss of local compactness. By the same diagonal procedure used in [Theorem 9.8](#), this alternative is realized by a terminal extraction of the original refined residual candidate, after passing to a subsequence and choosing the local pressure gauges supplied by Paper I. Loss of local compactness contradicts [Theorem 8.3](#). Diffuse exterior concentration places the retained compact velocity mass in the diffuse exterior alternative, contradicting [Theorem 8.8](#). Therefore alternative (ii) cannot occur for a refined residual family with retained local concentration. $\square$

Lemma 9.11 (Finite separated families produce indecomposable profiles or recurrent loops). *Let $A = \{\mathfrak{s}_1, \dots, \mathfrak{s}_J\}$, $J \geq 2$, be a finite retained concentration separated family for a refined residual candidate. Then either one of the terminal profiles (U_j, Π_j) is terminally indecomposable, or the family generates a finite recurrent loop of retained concentrating tail limits.*

Proof. Assume no (U_j, Π_j) is terminally indecomposable. By [Theorems 9.9](#) and [9.10](#), each (U_j, Π_j) has a retained concentrating tail limit which lifts to a recentering sequence of the original terminal sequence with retained local concentration.

If a lifted concentrating recentering sequence is not equivalent to one of the sequences already in the finite family, enlarge the family by adding this new sequence. This is still a finite retained concentration separated family because only one separated recentering sequence with retained local concentration has been added. Repeat the construction. At each finite stage there are two possibilities. Either an indecomposable profile appears, proving the first alternative, or every existing decomposable profile has a retained concentrating tail limit.

If the process returns to a previously occurring concentrating recentering sequence, the directed recentering relation contains a finite loop

$$\mathfrak{s}_{j_0} \rightarrow \mathfrak{s}_{j_1} \rightarrow \dots \rightarrow \mathfrak{s}_{j_q} = \mathfrak{s}_{j_0}.$$

This is the desired finite recurrent loop of retained concentrating tail limits.

It remains to explain why the construction cannot avoid both alternatives by producing concentrating recentering sequences escaping every finite stage. Such an escaping sequence of lifted concentrating recenterings is, by [Theorem 8.6](#), an exterior escaping concentration relative to each previously fixed finite family unless local compactness fails or the mass becomes diffuse below the unit concentration threshold. The latter two cases are loss of local compactness and diffuse exterior concentration, already excluded by [Theorem 9.10](#). The exterior case gives a retained compact CKN concentration outside every fixed finite family. Recenter at that exterior concentration. In that frame, all previously chosen finite family components leave every compact terminal cylinder by [Theorem 8.5](#); hence the retained compact density is located solely in the exterior escaping alternative relative to the preceding finite family. This contradicts the exterior case of [Theorem 8.8](#). Therefore indefinite escape without a return is impossible, and a finite recurrent loop must occur unless an indecomposable profile has already occurred. $\square$

Lemma 9.12 (Finite recurrent loops of retained tail limits are impossible). *No refined residual terminal extraction contains a finite recurrent loop of retained concentrating tail limits.*

Proof. Let

$$\mathfrak{s}_{j_0} \rightarrow \mathfrak{s}_{j_1} \rightarrow \cdots \rightarrow \mathfrak{s}_{j_q} = \mathfrak{s}_{j_0}$$

be such a loop, and write (U_ℓ, Π_ℓ) for the corresponding retained concentration terminal profiles. Each arrow means that $U_{\ell+1}$ is a retained concentrating tail limit of U_ℓ . Thus, for each ℓ , there are escaping tail recentering sequences whose recentered fields converge locally to $U_{\ell+1}$.

Let $\mathcal{K}$ be the closure, in the local smooth velocity topology and local $L^{3/2}$ pressure topology modulo gauges, of all profiles obtained from the finite set $\{U_0, \dots, U_{q-1}\}$ by time translations and covariant terminal translations. The boundedness inherited from the Type I Seregina extraction theorem and the local pressure compactness of Paper I give compactness of $\mathcal{K}$ on every fixed terminal cylinder; a diagonal argument gives compactness of $\mathcal{K}$. The set is invariant under the same time and covariant translations by construction. The recurrent-loop relations imply that each U_ℓ returns to $\mathcal{K}$ through tail limits, so the closed invariant subset of $\mathcal{K}$ generated by the loop is nonempty, compact, invariant, and contains the retained concentration profiles.

Choose a compact invariant subset $\mathcal{M}$ of this loop-generated set which is minimal among compact invariant subsets still containing the loop profiles. The invariant mean argument in [Theorem 7.12](#) applies to $\mathcal{M}$: all ingredients used there are local on compact cylinders, and the pressure term is justified by the invariant local pressure-gauge lemma [Theorem 7.11](#). Hence every element of $\mathcal{M}$, in particular every loop profile U_ℓ , is spatially and temporally constant. Applying the affine homogeneous symmetry [Theorem 7.5](#) removes this constant and sends each loop profile to the zero profile. This contradicts retained local concentration,

$$\mathcal{E}_1(U_\ell, \Pi_\ell; 0) \geq \varepsilon_*.$$

Therefore no finite recurrent loop of retained concentrating tail limits exists. $\square$

Theorem 9.13 (Finite separated retained concentration profile families are impossible). *No refined residual candidate admits a finite separated family of retained concentration profiles.*

Proof. Suppose, for contradiction, that a refined residual candidate admits a finite retained concentration separated family $A = \{\mathfrak{s}_1, \dots, \mathfrak{s}_J\}$, $J \geq 2$. By [Theorem 9.11](#), either some component of the family is terminally indecomposable or the family generates a finite recurrent loop of retained concentrating tail limits. The first alternative contradicts [Theorem 9.7](#). The second contradicts [Theorem 9.12](#). Hence the finite retained concentration separated family cannot exist. $\square$

Theorem 9.14 (Terminal local concentration theorem). *Every bounded centered Type I Seregina profile with retained local concentration outside the previously closed lower classes admits the local terminal concentration decomposition of [Theorem 8.6](#); loss of local compactness, local vanishing, exterior escape, and diffuse exterior concentration are excluded by [Theorem 8.9](#); finite separated retained concentration profile families are excluded by [Theorem 9.13](#); every remaining single retained concentration terminal profile is terminally indecomposable by [Theorem 9.3](#); retained indecomposable profiles have a backward sequence of finite critical norm by [Theorem 9.4](#); and retained concentration terminal profiles inherit mildness by [Theorem 9.5](#).*

Proof. This is the assembly of the local results proved in [Sections 8 and 9](#). The local concentration decomposition is [Theorem 8.6](#). The local-vanishing, exterior-escape, diffuse-exterior-concentration, and local-compactness-failure alternatives are excluded by [Theorem 8.9](#). The finite separated concentration profile-family alternative is empty by [Theorem 9.13](#). Therefore any retained concentration terminal profile that survives the ordered lower class removals is single. By

[Theorem 9.3](#) it is terminally indecomposable, and then [Theorems 9.4](#) and [9.5](#) give the sequence- L^3 and mildness conclusions. All estimates used in this assembly are localized to fixed terminal cylinders, except for the final sequence L^3 conclusion, which is derived by contradiction from terminal indecomposability and is not assumed as a global bound. $\square$

10. CLOSURE OF THE GENERIC TERMINAL RESIDUAL

Definition 10.1 (Generic residual profile). A generic residual profile is a smooth bounded normalized Seregin ancient solution of [\(1.2\)](#) which carries persistent compact velocity L^3 -mass and belongs to none of the previously closed alternatives, none of $\mathcal{C}_{\text{ax}}, \mathcal{C}_{\text{rot}}, \mathcal{C}_{\text{stat}}^{\text{ret}}$, and none of the structured or tight classes removed earlier.

Theorem 10.2 (Closure of the generic residual class by local concentration analysis). *One has*

$$\mathcal{C}_{\text{gen}} = \emptyset.$$

Proof. Suppose, for contradiction, that $V \in \mathcal{C}_{\text{gen}}$. By the tail geometry developed above, constant tails may be removed by the exact affine symmetry and nonconstant recurrent profile tail cores collapse to affine constants. Thus the remaining representative may be taken to be tail-normalized without leaving the centered equation or the residual decomposition. The inherited compact velocity L^3 -mass remains positive.

Apply the terminal local concentration decomposition to a terminal extraction of this tail-normalized candidate. Local vanishing, exterior escape, diffuse exterior concentration, and loss of local compactness cannot be the sole locations of the persistent compact local concentration in the generic residual, by [Theorem 8.9](#). Therefore the retained concentration is located either in a finite separated family of retained concentration profiles or in a single retained concentration terminal profile.

The finite separated-family case contradicts [Theorem 9.13](#). In the single-profile case, [Theorem 9.3](#) makes the terminal profile terminally indecomposable. Then [Theorem 9.7](#) excludes it. Both possible locations of the retained compact velocity mass are impossible. Hence no generic residual profile exists. $\square$

11. FINAL ASSEMBLY

The class closures above form the following dependency chain. The rotational relative-equilibrium arrow is the localized Coriolis-flux argument; the terminal local concentration, no finite separated-family, and indecomposability arrows are proved in this paper from local estimates:

$$\begin{aligned} \mathcal{C}_{\text{ax}} &\xrightarrow{\text{r-Liouville}} \emptyset, \\ \mathcal{C}_{\text{rot}} &\xrightarrow{\text{local flux}} \emptyset, \\ \mathcal{C}_{\text{stat}}^{\text{ret}} &\xrightarrow{\text{terminal concentration}} \emptyset, \\ \mathcal{C}_{\text{gen}} &\xrightarrow{\text{tail normalization, finite separation, sequence-}L^3} \emptyset. \end{aligned}$$

The positive local velocity mass inherited by every normalized Seregin limit is the common contradiction target in all four classes.

Proof of [Theorem 1.5](#), revisited. The detailed reductions have now been given in the four class sections. The axisymmetric bounded-circulation class is empty unconditionally by [Theorem 3.3](#). The rotational relative-equilibrium class is empty by the local Coriolis-flux and recentered concentration analysis in [Theorem 4.8](#). The degenerate stationary-hull class with retained local concentration is empty by the terminal local concentration theorem, through [Theorem 6.6](#). The

generic terminal residual is empty by the finite separated-family and indecomposability argument in [Theorem 10.2](#). Since the refined decomposition exhausts the residual class by definition, these four local exclusions imply that the residual class is empty. $\square$

Remark 11.1 (What is imported and what is proved here). The paper imports from Paper 0 only local positive concentration and the local Type I/Type II dichotomy, imports from Paper I only the Type I Seregin extraction and residual criterion, imports from Paper II only the results proved there, and imports from Paper III only the closed structured alternatives. The terminal finite separated-family exclusion is not imported: it is implemented above by local concentration analysis. The rotational relative-equilibrium class is not closed by a global Coriolis–Leray Liouville theorem for bounded profiles; it is closed by the localized flux identity and the same recentered local concentration analysis.

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